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3D Geometry

JEE Main Mathematics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 157 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Let the foot of perpendicular from the point $(\lambda, 2, 3)$ on the line $\frac{x-4}{1} = \frac{y-9}{2} = \frac{z-5}{1}$ be the point ($1, \mu, 2$). Then the distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+4}{6}$ and $\frac{x-\lambda}{2} = \frac{y-\mu}{3} = \frac{z+5}{6}$ is equal to :

(A) $\frac{12}{7}$ (B) $\frac{\sqrt{145}}{7}$ (C) $\frac{\sqrt{146}}{7}$ (D) $\frac{\sqrt{143}}{7}$

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

The shortest distance between the lines $\frac{x-4}{1} = \frac{y-3}{2} = \frac{z-2}{-3}$ and $\frac{x+2}{2} = \frac{y-6}{4} = \frac{z-5}{-5}$ is:

(A) $\frac{5\sqrt{6}}{6}$ (B) $2\sqrt{5}$ (C) $3\sqrt{5}$ (D) $4\sqrt{5}$

Open solution

Q3. JEE Main 2026 (Online) 6th April Morning Shift

Let the image of the point $\mathrm{P}(1, 6, a)$ in the line $\mathrm{L} : \frac{x}{1} = \frac{y-1}{2} = \frac{z-a+1}{b}, b > 0$, be $\left(\frac{a}{3}, 0, a+c\right)$. If $\mathrm{S}(\alpha, \beta, \gamma), \alpha > 0$, is the point on L such that the distance of S from the foot of perpendicular from the point P on L is $2\sqrt{14}$, then $\alpha + \beta + \gamma$ is equal to:

(A) 19 (B) 20 (C) 21 (D) 22

Open solution

Q4. JEE Main 2026 (Online) 6th April Morning Shift

Let a line L be perpendicular to both the lines $\mathrm{L}_1 : \frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and $\mathrm{L}_2 : \frac{x-2}{1} = \frac{y-4}{4} = \frac{z-6}{7}$.

If θ is the acute angle between the lines L and $\mathrm{L}_3 : \frac{x-\frac{8}{7}}{2} = \frac{y-\frac{4}{7}}{1} = \frac{z}{2}$, then $\tan\theta$ is equal to:

(A) $\frac{3}{2}\sqrt{2}$ (B) $\frac{5}{2}\sqrt{2}$ (C) $\frac{5}{3}\sqrt{2}$ (D) $\frac{4}{3}\sqrt{2}$

Open solution

Q5. JEE Main 2026 (Online) 5th April Evening Shift

Let a triangle PQR be such that P and Q lie on the line $\frac{x+3}{8} = \frac{y-4}{2} = \frac{z+1}{2}$ and are at a distance of 6 units from $R(1, 2, 3)$. If (α, β, γ) is the centroid of ΔPQR, then $\alpha + \beta + \gamma$ is equal to :

(A) 4 (B) 5 (C) 6 (D) 8

Open solution

Q6. JEE Main 2026 (Online) 5th April Evening Shift

If the distance of the point $(a, 2, 5)$ from the image of the point $(1, 2, 7)$ in the line $\frac{x}{1} = \frac{y-1}{1} = \frac{z-2}{2}$ is 4 , then the sum of all possible values of a is equal to :

(A) 11 (B) 9 (C) 6 (D) 4

Open solution

Q7. JEE Main 2026 (Online) 5th April Morning Shift

The square of the distance of the point $\mathrm{P}(5, 6, 7)$ from the line $\frac{x-2}{2} = \frac{y-5}{3} = \frac{z-2}{4}$ is equal to:

(A) 3 (B) 5 (C) 6 (D) 8

Open solution

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Q8. JEE Main 2026 (Online) 5th April Morning Shift

The square of the distance of the point of intersection of the lines $\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(a\hat{i} - \hat{j}), a \neq 0$ and $\vec{r} = (4\hat{i} - \hat{k}) + \mu(2\hat{i} + a\hat{k})$ from the origin is :

(A) 5 (B) 10 (C) 17 (D) 26

Open solution

Q9. JEE Main 2026 (Online) 4th April Evening Shift

The shortest distance between the lines

$\vec{r} = \left(\frac{1}{3}\hat{\mathrm{i}} + 2\hat{\mathrm{j}} + \frac{8}{3}\hat{\mathrm{k}}\right) + \lambda(2\hat{\mathrm{i}} - 5\hat{\mathrm{j}} + 6\hat{\mathrm{k}})$

and $\vec{r} = \left(-\frac{2}{3}\hat{\mathrm{i}} - \frac{1}{3}\hat{\mathrm{k}}\right) + \mu(\hat{\mathrm{j}} - \hat{\mathrm{k}}), \lambda, \mu \in \mathbb{R}$, is:

(A) $\sqrt{5}$ (B) 3 (C) $2\sqrt{3}$ (D) $\sqrt{15}$

Open solution

Q10. JEE Main 2026 (Online) 4th April Evening Shift

If $\left(2\alpha + 1, \alpha^2 - 3\alpha, \frac{\alpha-1}{2}\right)$ is the image of $(\alpha, 2\alpha, 1)$ in the line $\frac{x-2}{3} = \frac{y-1}{2} = \frac{z}{1}$, then the possible value(s) of α is (are)

(A) Only 3 (B) Only 3 and - 1

(C) Only 3, $\frac{1}{4}$ and -1 (D) Only 3 and $\frac{1}{4}$

Open solution

Q11. JEE Main 2026 (Online) 4th April Morning Shift

A line with direction ratios $1, -1, 2$ intersects the lines $\frac{x}{2} = \frac{y}{3} = \frac{z+1}{3}$ and $\frac{x+1}{-1} = \frac{y-2}{1} = \frac{z}{4}$ at the points P and Q , respectively. If the length of the line segment PQ is α, then $225\alpha^2$ is equal to:

(A) 1024 (B) 1014 (C) 1104 (D) 1204

Open solution

Q12. JEE Main 2026 (Online) 4th April Morning Shift

The square of the distance of the point $(-2, -8, 6)$ from the line $\frac{x-1}{1} = \frac{y-1}{2} = \frac{z}{-1}$ along the line $\frac{x+5}{1} = \frac{y+5}{-1} = \frac{z}{2}$ is equal to:

(A) 3 (B) 6 (C) 8 (D) 12

Open solution

Q13. JEE Main 2026 (Online) 2nd April Evening Shift

Let the point A be the foot of perpendicular drawn from the point P$(a, b, 0)$ on the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-\alpha}{3}$.

If the midpoint of the line segment PA is $\left(0, \frac{3}{4}, -\frac{1}{4}\right)$, then the value of $a^2 + b^2 + \alpha^2$ is equal to :

(A) 1 (B) 2 (C) 6 (D) 9

Open solution

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Q14. JEE Main 2026 (Online) 2nd April Morning Shift

If the point of intersection of the lines $\frac{x+1}{3}=\frac{y+a}{5}=\frac{z+b+1}{7}$ and $\frac{x-2}{1}=\frac{y-b}{4}=\frac{z-2a}{7}$ lies on xy-plane, then the value of $a+b$ is:

(A) 2 (B) 5 (C) 7 (D) 9

Open solution

Q15. JEE Main 2026 (Online) 2nd April Morning Shift

Let a line L passing through the point (1, 1, 1) be perpendicular to both the vectors $2\hat{i}+2\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}+2\hat{k}$. If $P(a,b,c)$ is the foot of perpendicular from the origin on the line L, then the value of $34(a+b+c)$ is :

(A) 50 (B) 80 (C) 100 (D) 120

Open solution

Q16. JEE Main 2026 (Online) 28th January Evening Shift

Let Q(a, b, c) be the image of the point P(3, 2, 1) in the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$. Then the distance of Q from the line $\frac{x-9}{3}=\frac{y-9}{2}=\frac{z-5}{-2}$ is

(A) 8 (B) 7 (C) 6 (D) 5

Open solution

Q17. JEE Main 2026 (Online) 28th January Morning Shift

If the distances of the point $(1,2,a)$ from the line $\frac{x-1}{1}=\frac{y}{2}=\frac{z-1}{1}$ along the lines $L_1:\frac{x-1}{3}=\frac{y-2}{4}=\frac{z-a}{b}$ and $L_2:\frac{x-1}{1}=\frac{y-2}{4}=\frac{z-a}{c}$ are equal, then $a+b+c$ is equal to

(A) 4 (B) 6 (C) 7 (D) 5

Open solution

Q18. JEE Main 2026 (Online) 24th January Evening Shift

The sum of all values of α, for which the shortest distance between the lines $\frac{x+1}{\alpha}=\frac{y-2}{-1}=\frac{z-4}{-\alpha}$ and $\frac{x}{\alpha}=\frac{y-1}{2}=\frac{z-1}{2\alpha}$ is $\sqrt{2}$, is

(A) -6 (B) -8 (C) 8 (D) 6

Open solution

Q19. JEE Main 2026 (Online) 24th January Morning Shift

Let the lines $L_1:\vec{r}=\hat{i}+2\hat{j}+3\hat{k}+\lambda(2\hat{i}+3\hat{j}+4\hat{k}),\lambda\in\mathbb{R}$ and $L_2:\vec{r}=(4\hat{i}+\hat{j})+\mu(5\hat{i}+2\hat{j}+\hat{k}),\mu\in\mathbb{R}$, intersect at the point R . Let P and Q be the points lying on lines L_1 and L_2, respectively, such that $|\overrightarrow{PR}|=\sqrt{29}$ and $|\overrightarrow{PQ}|=\sqrt{\frac{47}{3}}$. If the point P lies in the first octant, then $27(QR)^2$ is equal to

(A) 360 (B) 348 (C) 320 (D) 340

Open solution

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Q20. JEE Main 2026 (Online) 23rd January Morning Shift

Let the direction cosines of two lines satisfy the equations : $4l + m - n = 0$ and $2mn + 10nl + 3lm = 0$.

Then the cosine of the acute angle between these lines is :

(A) $\frac{10}{7\sqrt{38}}$ (B) $\frac{10}{\sqrt{38}}$ (C) $\frac{10}{3\sqrt{38}}$ (D) $\frac{20}{3\sqrt{38}}$

Open solution

Q21. JEE Main 2026 (Online) 23rd January Morning Shift

The vertices B and C of a triangle ABC lie on the line $\frac{x}{1} = \frac{1-y}{-2} = \frac{z-2}{3}$. The coordinates of A and B are $(1, 6, 3)$ and $(4, 9, \alpha)$ respectively and C is at a distance of 10 units from B. The area (in sq. units) of $\triangle ABC$ is :

(A) $20\sqrt{13}$ (B) $5\sqrt{13}$ (C) $15\sqrt{13}$ (D) $10\sqrt{13}$

Open solution

Q22. JEE Main 2026 (Online) 22nd January Evening Shift

Let L be the line $\frac{x+1}{2} = \frac{y+1}{3} = \frac{z+3}{6}$ and let S be the set of all points $(\mathrm{a}, \mathrm{b}, \mathrm{c})$ on L , whose distance from the line $\frac{x+1}{2} = \frac{y+1}{3} = \frac{z-9}{0}$ along the line L is 7 . Then $\sum_{(a,b,c)\in S} (a+b+c)$ is equal to :

(A) 28 (B) 6 (C) 40 (D) 34

Open solution

Q23. JEE Main 2026 (Online) 22nd January Morning Shift

Let $\mathrm{P}(\alpha, \beta, \gamma)$ be the point on the line $\frac{x-1}{2} = \frac{y+1}{-3} = z$ at a distance $4\sqrt{14}$ from the point $(1, -1, 0)$ and nearer to the origin. Then the shortest distance, between the lines $\frac{x-\alpha}{1} = \frac{y-\beta}{2} = \frac{z-\gamma}{3}$ and $\frac{x+5}{2} = \frac{y-10}{1} = \frac{z-3}{1}$, is equal to

(A) $4\sqrt{\frac{7}{5}}$ (B) $7\sqrt{\frac{5}{4}}$ (C) $4\sqrt{\frac{5}{7}}$ (D) $2\sqrt{\frac{7}{4}}$

Open solution

Q24. JEE Main 2026 (Online) 22nd January Morning Shift

If the image of the point $\mathrm{P}(1, 2, a)$ in the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{7-z}{2}$ is $\mathrm{Q}(5, b, \mathrm{c})$, then $a^2 + b^2 + c^2$ is equal to

(A) 298 (B) 264 (C) 293 (D) 283

Open solution

Q25. JEE Main 2026 (Online) 21st January Evening Shift

Let the line L pass through the point $(-3, 5, 2)$ and make equal angles with the positive coordinate axes. If the distance of L from the point $(-2, r, 1)$ is $\sqrt{\frac{14}{3}}$, then the sum of all possible values of r is :

(A) 16 (B) 12 (C) 6 (D) 10

Open solution

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Q26. JEE Main 2026 (Online) 21st January Evening Shift

Let the line L_1 be parallel to the vector $-3\hat{i}+2\hat{j}+4\hat{k}$ and pass through the point $(2,6,7)$, and the line L_2 be parallel to the vector $2\hat{i}+\hat{j}+3\hat{k}$ and pass through the point $(4,3,5)$. If the line L_3 is parallel to the vector $-3\hat{i}+5\hat{j}+16\hat{k}$ and intersects the lines L_1 and L_2 at the points C and D, respectively, then $\left|\overrightarrow{CD}\right|^2$ is equal to:

Open solution

(A) 290 (B) 171 (C) 89 (D) 312

Q27. JEE Main 2026 (Online) 21st January Morning Shift

Let (α,β,γ) be the co-ordinates of the foot of the perpendicular drawn from the point $(5,4,2)$ on the line $\vec{r}=(-\hat{i}+3\hat{j}+\hat{k})+\lambda(2\hat{i}+3\hat{j}-\hat{k})$.

Then the length of the projection of the vector $\alpha\hat{i}+\beta\hat{j}+\gamma\hat{k}$ on the vector $6\hat{i}+2\hat{j}+3\hat{k}$ is :

Open solution

(A) $\frac{18}{7}$ (B) $\frac{15}{7}$ (C) 4 (D) 3

Q28. JEE Main 2025 (Online) 8th April Evening Shift

Let the values of λ for which the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-\lambda}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ is $\frac{1}{\sqrt{6}}$ be λ_1 and λ_2. Then the radius of the circle passing through the points $(0,0),(\lambda_1,\lambda_2)$ and (λ_2,λ_1) is

Open solution

(A) 3 (B) $\frac{5\sqrt{2}}{3}$ (C) $\frac{\sqrt{2}}{3}$ (D) 4

Q29. JEE Main 2025 (Online) 7th April Evening Shift

If the equation of the line passing through the point $\left(0,-\frac{1}{2},0\right)$ and perpendicular to the lines $\vec{r}=\lambda\left(\hat{i}+a\hat{j}+b\hat{k}\right)$ and $\vec{r}=\left(\hat{i}-\hat{j}-6\hat{k}\right)+\mu\left(-b\hat{i}+a\hat{j}+5\hat{k}\right)$ is $\frac{x-1}{-2}=\frac{y+4}{d}=\frac{z-c}{-4}$, then $a+b+c+d$ is equal to :

Open solution

(A) 13 (B) 14 (C) 12 (D) 10

Q30. JEE Main 2025 (Online) 7th April Evening Shift

Consider the lines $\mathbf{L_1}$: x - 1 = y - 2 = z and $\mathbf{L_2}$: x - 2 = y = z - 1. Let the feet of the perpendiculars from the point P(5, 1, -3) on the lines $\mathbf{L_1}$ and $\mathbf{L_2}$ be Q and R respectively. If the area of the triangle PQR is A, then $4A^2$ is equal to :

Open solution

(A) 151 (B) 147 (C) 139 (D) 143

Q31. JEE Main 2025 (Online) 7th April Morning Shift

Let the line L pass through $(1,1,1)$ and intersect the lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-1}{4}$ and $\frac{x-3}{1}=\frac{y-4}{2}=\frac{z}{1}$. Then, which of the following points lies on the line L ?

Open solution

(A) $(7,15,13)$ (B) $(4,22,7)$ (C) $(10,-29,-50)$ (D) $(5,4,3)$

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Q32. JEE Main 2025 (Online) 7th April Morning Shift

If the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x}{1}=\frac{y}{\alpha}=\frac{z-5}{1}$ is $\frac{5}{\sqrt{6}}$, then the sum of all possible values of α is

(A) $\frac{3}{2}$ (B) 3 (C) -3 (D) $-\frac{3}{2}$

Open solution

Q33. JEE Main 2025 (Online) 4th April Evening Shift

Let A be the point of intersection of the lines $\mathrm{L}_1: \frac{x-7}{1}=\frac{y-5}{0}=\frac{z-3}{-1}$ and $\mathrm{L}_2: \frac{x-1}{3}=\frac{y+3}{4}=\frac{z+7}{5}$. Let B and C be the points on the lines L_1 and L_2 respectively such that $AB=AC=\sqrt{15}$. Then the square of the area of the triangle ABC is :

(A) 63 (B) 57 (C) 60 (D) 54

Open solution

Q34. JEE Main 2025 (Online) 4th April Evening Shift

Let the values of p , for which the shortest distance between the lines $\frac{x+1}{3}=\frac{y}{4}=\frac{z}{5}$ and $\vec{r}=(\mathrm{p}\hat{i}+2\hat{j}+\hat{k})+\lambda(2\hat{i}+3\hat{j}+4\hat{k})$ is $\frac{1}{\sqrt{6}}$, be $\mathrm{a}, \mathrm{b}, (\mathrm{a}<\mathrm{b})$. Then the length of the latus rectum of the ellipse $\frac{x^2}{\mathrm{a}^2}+\frac{y^2}{\mathrm{b}^2}=1$ is :

(A) $\frac{3}{2}$ (B) 9 (C) 18 (D) $\frac{2}{3}$

Open solution

Q35. JEE Main 2025 (Online) 4th April Morning Shift

Let the shortest distance between the lines $\frac{x-3}{3}=\frac{y-\alpha}{-1}=\frac{z-3}{1}$ and $\frac{x+3}{-3}=\frac{y+7}{2}=\frac{z-\beta}{4}$ be $3\sqrt{30}$. Then the positive value of $5\alpha+\beta$ is

(A) 42 (B) 40 (C) 48 (D) 46

Open solution

Q36. JEE Main 2025 (Online) 4th April Morning Shift

Let A and B be two distinct points on the line $L: \frac{x-6}{3}=\frac{y-7}{2}=\frac{z-7}{-2}$. Both A and B are at a distance $2\sqrt{17}$ from the foot of perpendicular drawn from the point $(1,2,3)$ on the line L. If O is the origin, then $\overrightarrow{OA}\cdot\overrightarrow{OB}$ is equal to

(A) 49 (B) 21 (C) 47 (D) 62

Open solution

Q37. JEE Main 2025 (Online) 3rd April Evening Shift

Each of the angles β and γ that a given line makes with the positive y - and z-axes, respectively, is half of the angle that this line makes with the positive x-axes. Then the sum of all possible values of the angle β is

(A) $\frac{\pi}{2}$ (B) π (C) $\frac{3\pi}{4}$ (D) $\frac{3\pi}{2}$

Open solution

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Q38. JEE Main 2025 (Online) 3rd April Evening Shift

The distance of the point $(7, 10, 11)$ from the line $\frac{x-4}{1} = \frac{y-4}{0} = \frac{z-2}{3}$ along the line $\frac{x-9}{2} = \frac{y-13}{3} = \frac{z-17}{6}$ is

(A) 16 (B) 12 (C) 18 (D) 14

Open solution

Q39. JEE Main 2025 (Online) 3rd April Morning Shift

Let a line passing through the point $(4, 1, 0)$ intersect the line $\mathrm{L}_1 : \frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ at the point $A(\alpha, \beta, \gamma)$ and the line $\mathrm{L}_2 : x - 6 = y = -z + 4$ at the point $B(a, b, c)$. Then $\begin{vmatrix} 1 & 0 & 1 \\ \alpha & \beta & \gamma \\ a & b & c \end{vmatrix}$ is equal to

(A) 16 (B) 6 (C) 8 (D) 12

Open solution

Q40. JEE Main 2025 (Online) 3rd April Morning Shift

Line L_1 passes through the point $(1, 2, 3)$ and is parallel to z-axis. Line L_2 passes through the point $(\lambda, 5, 6)$ and is parallel to y-axis. Let for $\lambda = \lambda_1, \lambda_2, \lambda_2 < \lambda_1$, the shortest distance between the two lines be 3 . Then the square of the distance of the point $(\lambda_1, \lambda_2, 7)$ from the line L_1 is

(A) 25 (B) 32 (C) 40 (D) 37

Open solution

Q41. JEE Main 2025 (Online) 2nd April Evening Shift

If the image of the point $\mathrm{P}(1, 0, 3)$ in the line joining the points $\mathrm{A}(4, 7, 1)$ and $\mathrm{B}(3, 5, 3)$ is $Q(\alpha, \beta, \gamma)$, then $\alpha + \beta + \gamma$ is equal to :

(A) $\frac{46}{3}$ (B) 18 (C) 13 (D) $\frac{47}{3}$

Open solution

Q42. JEE Main 2025 (Online) 2nd April Evening Shift

The line L_1 is parallel to the vector $\vec{\mathrm{a}} = -3\hat{i} + 2\hat{j} + 4\hat{k}$ and passes through the point $(7, 6, 2)$ and the line L_2 is parallel to the vector $\vec{\mathrm{b}} = 2\hat{i} + \hat{j} + 3\hat{k}$ and passes through the point $(5, 3, 4)$. The shortest distance between the lines L_1 and L_2 is :

(A) $\frac{23}{\sqrt{38}}$ (B) $\frac{21}{\sqrt{38}}$ (C) $\frac{23}{\sqrt{57}}$ (D) $\frac{21}{\sqrt{57}}$

Open solution

Q43. JEE Main 2025 (Online) 2nd April Morning Shift

Let the vertices Q and R of the triangle PQR lie on the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$, $\mathrm{QR} = 5$ and the coordinates of the point P be $(0, 2, 3)$. If the area of the triangle PQR is $\frac{m}{n}$ then :

(A) $2\,\mathrm{m} - 5\sqrt{21}\mathrm{n} = 0$ (B) $\mathrm{m} - 5\sqrt{21}\mathrm{n} = 0$

(C) $5\,\mathrm{m} - 21\sqrt{2}\mathrm{n} = 0$ (D) $5\,\mathrm{m} - 2\sqrt{21}\mathrm{n} = 0$

Open solution

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Q44. JEE Main 2025 (Online) 2nd April Morning Shift

Let $ABCD$ be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the areas of the triangles ABC, ACD and ADB be 5,6 and 7 square units respectively. Then the area (in square units) of the $\triangle BCD$ is equal to :

(A) $\sqrt{110}$ (B) 12 (C) $\sqrt{340}$ (D) $7\sqrt{3}$

Open solution

Q45. JEE Main 2025 (Online) 29th January Evening Shift

Let a straight line L pass through the point $P(2,-1,3)$ and be perpendicular to the lines $\frac{x-1}{2}=\frac{y+1}{1}=\frac{z-3}{-2}$ and $\frac{x-3}{1}=\frac{y-2}{3}=\frac{z+2}{4}$. If the line L intersects the yz-plane at the point Q, then the distance between the points P and Q is:

(A) $\sqrt{10}$ (B) 2 (C) $2\sqrt{3}$ (D) 3

Open solution

Q46. JEE Main 2025 (Online) 29th January Evening Shift

Let P be the foot of the perpendicular from the point $(1,2,2)$ on the line L : $\frac{x-1}{1}=\frac{y+1}{-1}=\frac{z-2}{2}$. Let the line $\vec{r}=(-\hat{i}+\hat{j}-2\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k}), \lambda \in \mathbf{R}$, intersect the line L at Q . Then $2(\mathrm{PQ})^2$ is equal to :

(A) 25 (B) 27 (C) 19 (D) 29

Open solution

Q47. JEE Main 2025 (Online) 29th January Morning Shift

Let L_1 : $\frac{x-1}{1}=\frac{y-2}{-1}=\frac{z-1}{2}$ and L_2 : $\frac{x+1}{-1}=\frac{y-2}{2}=\frac{z}{1}$ be two lines.

Let L_3 be a line passing through the point (α,β,γ) and be perpendicular to both L_1 and L_2. If L_3 intersects L_1, then $|5\alpha-11\beta-8\gamma|$ equals :

(A) 25 (B) 20 (C) 16 (D) 18

Open solution

Q48. JEE Main 2025 (Online) 29th January Morning Shift

Let $\vec{a}=\hat{i}+2\hat{j}+\hat{k}$ and $\vec{b}=2\hat{i}+7\hat{j}+3\hat{k}$. Let $\mathrm{L}_1 : \overrightarrow{\mathrm{r}}=(-\hat{i}+2\hat{j}+\hat{k})+\lambda\vec{a}, \lambda \in \mathbf{R}$ and $\mathrm{L}_2 : \overrightarrow{\mathrm{r}}=(\hat{j}+\hat{k})+\mu\vec{b}, \mu \in \mathrm{R}$ be two lines. If the line L_3 passes through the point of intersection of L_1 and L_2 and is parallel to $\vec{a}+\vec{b}$, then L_3 passes through the point :

(A) $(-1,-1,1)$ (B) $(2,8,5)$ (C) $(8,26,12)$ (D) $(5,17,4)$

Open solution

Q49. JEE Main 2025 (Online) 28th January Evening Shift

The square of the distance of the point $(\frac{15}{7},\frac{32}{7},7)$ from the line $\frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ in the direction of the vector $\hat{i}+4\hat{j}+7\hat{k}$ is:

(A) 66 (B) 54 (C) 41 (D) 44

Open solution

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Q50. JEE Main 2025 (Online) 28th January Morning Shift

Let $\mathrm{A}(x, y, z)$ be a point in xy-plane, which is equidistant from three points $(0, 3, 2), (2, 0, 3)$ and $(0, 0, 1)$.

Let $\mathrm{B} = (1, 4, -1)$ and $\mathrm{C} = (2, 0, -2)$. Then among the statements

(S1) : $\triangle\mathrm{ABC}$ is an isosceles right angled triangle, and

(S2) : the area of $\triangle\mathrm{ABC}$ is $\frac{9\sqrt{2}}{2}$,

(A) both are false
(B) only (S2) is true
(C) only (S1) is true
(D) both are true

Open solution

Q51. JEE Main 2025 (Online) 28th January Morning Shift

If the image of the point $(4, 4, 3)$ in the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-1}{3}$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to

(A) 12 (B) 9 (C) 7 (D) 8

Open solution

Q52. JEE Main 2025 (Online) 24th January Morning Shift

Let in a $\triangle ABC$, the length of the side AC be 6 , the vertex B be $(1, 2, 3)$ and the vertices A, C lie on the line $\frac{x-6}{3} = \frac{y-7}{2} = \frac{z-7}{-2}$. Then the area (in sq. units) of $\triangle ABC$ is:

(A) 42 (B) 17 (C) 56 (D) 21

Open solution

Q53. JEE Main 2025 (Online) 24th January Morning Shift

Let the line passing through the point $(-1, 2, 1)$ and parallel to the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z}{4}$ intersect the line $\frac{x+2}{3} = \frac{y-3}{2} = \frac{z-4}{1}$ at the point P. Then the distance of P from the point $Q(4, -5, 1)$ is

(A) $5\sqrt{6}$ (B) 5 (C) $5\sqrt{5}$ (D) 10

Open solution

Q54. JEE Main 2025 (Online) 23rd January Evening Shift

If the square of the shortest distance between the lines $\frac{x-2}{1} = \frac{y-1}{2} = \frac{z+3}{-3}$ and $\frac{x+1}{2} = \frac{y+3}{4} = \frac{z+5}{-5}$ is $\frac{m}{n}$, where m, n are coprime numbers, then $m + n$ is equal to :

(A) 14 (B) 6 (C) 21 (D) 9

Open solution

Q55. JEE Main 2025 (Online) 23rd January Evening Shift

The distance of the line $\frac{x-2}{2} = \frac{y-6}{3} = \frac{z-3}{4}$ from the point $(1, 4, 0)$ along the line $\frac{x}{1} = \frac{y-2}{2} = \frac{z+3}{3}$ is :

(A) $\sqrt{17}$ (B) $\sqrt{13}$ (C) $\sqrt{15}$ (D) $\sqrt{14}$

Open solution

Q56. JEE Main 2025 (Online) 23rd January Morning Shift

Let P be the foot of the perpendicular from the point $\mathrm{Q}(10, -3, -1)$ on the line $\frac{x-3}{7} = \frac{y-2}{-1} = \frac{z+1}{-2}$. Then the area of the right angled triangle PQR, where R is the point $(3, -2, 1)$, is

(A) $\sqrt{30}$ (B) $9\sqrt{15}$ (C) $3\sqrt{30}$ (D) $8\sqrt{15}$

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Q57. JEE Main 2025 (Online) 22nd January Evening Shift

Let a line pass through two distinct points $P(-2,-1,3)$ and Q, and be parallel to the vector $3\hat{i}+2\hat{j}+2\hat{k}$. If the distance of the point Q from the point $\mathrm{R}(1,3,3)$ is 5 , then the square of the area of $\triangle PQR$ is equal to :

(A) 148 (B) 144 (C) 136 (D) 140

Open solution

Q58. JEE Main 2025 (Online) 22nd January Evening Shift

The perpendicular distance, of the line $\frac{x-1}{2}=\frac{y+2}{-1}=\frac{z+3}{2}$ from the point $\mathrm{P}(2,-10,1)$, is :

(A) 6 (B) $4\sqrt{3}$ (C) $3\sqrt{5}$ (D) $5\sqrt{2}$

Open solution

Q59. JEE Main 2025 (Online) 22nd January Morning Shift

Let $\mathrm{L}_1: \frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\mathrm{L}_2: \frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ be two lines. Then which of the following points lies on the line of the shortest distance between L_1 and L_2 ?

(A) $\left(\frac{14}{3},-3,\frac{22}{3}\right)$ (B) $\left(2,3,\frac{1}{3}\right)$
(C) $\left(\frac{8}{3},-1,\frac{1}{3}\right)$ (D) $\left(-\frac{5}{3},-7,1\right)$

Open solution

Q60. JEE Main 2024 (Online) 9th April Evening Shift

Consider the line L passing through the points $(1,2,3)$ and $(2,3,5)$. The distance of the point $\left(\frac{11}{3},\frac{11}{3},\frac{19}{3}\right)$ from the line L along the line $\frac{3x-11}{2}=\frac{3y-11}{1}=\frac{3z-19}{2}$ is equal to

(A) 6 (B) 3 (C) 5 (D) 4

Open solution

Q61. JEE Main 2024 (Online) 9th April Morning Shift

The shortest distance between the lines $\frac{x-3}{4}=\frac{y+7}{-11}=\frac{z-1}{5}$ and $\frac{x-5}{3}=\frac{y-9}{-6}=\frac{z+2}{1}$ is:

(A) $\frac{185}{\sqrt{563}}$ (B) $\frac{187}{\sqrt{563}}$ (C) $\frac{178}{\sqrt{563}}$ (D) $\frac{179}{\sqrt{563}}$

Open solution

Q62. JEE Main 2024 (Online) 9th April Morning Shift

Let the line L intersect the lines $x-2=-y=z-1, 2(x+1)=2(y-1)=z+1$ and be parallel to the line $\frac{x-2}{3}=\frac{y-1}{1}=\frac{z-2}{2}$. Then which of the following points lies on L ?

(A) $\left(-\frac{1}{3},1,-1\right)$ (B) $\left(-\frac{1}{3},-1,1\right)$ (C) $\left(-\frac{1}{3},-1,-1\right)$ (D) $\left(-\frac{1}{3},1,1\right)$

Open solution

Q63. JEE Main 2024 (Online) 8th April Evening Shift

If the shortest distance between the lines $\frac{x-\lambda}{2}=\frac{y-4}{3}=\frac{z-3}{4}$ and $\frac{x-2}{4}=\frac{y-4}{6}=\frac{z-7}{8}$ is $\frac{13}{\sqrt{29}}$, then a value of λ is :

(A) $\frac{13}{25}$ (B) 1 (C) -1 (D) $-\frac{13}{25}$

Open solution

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Q64. JEE Main 2024 (Online) 8th April Morning Shift

Let $P(x, y, z)$ be a point in the first octant, whose projection in the xy-plane is the point Q. Let $OP = \gamma$; the angle between OQ and the positive x-axis be θ; and the angle between OP and the positive z-axis be ϕ, where O is the origin. Then the distance of P from the x-axis is

(A) $\gamma\sqrt{1 - \sin^2\phi\cos^2\theta}$

(B) $\gamma\sqrt{1 + \cos^2\theta\sin^2\phi}$

(C) $\gamma\sqrt{1 + \cos^2\phi\sin^2\theta}$

(D) $\gamma\sqrt{1 - \sin^2\theta\cos^2\phi}$

Open solution

Q65. JEE Main 2024 (Online) 8th April Morning Shift

If the shortest distance between the lines

$L_1 : \vec{r} = (2+\lambda)\hat{i} + (1-3\lambda)\hat{j} + (3+4\lambda)\hat{k}, \quad \lambda \in \mathbb{R}$

$L_2 : \vec{r} = 2(1+\mu)\hat{i} + 3(1+\mu)\hat{j} + (5+\mu)\hat{k}, \quad \mu \in \mathbb{R}$

is $\frac{m}{\sqrt{n}}$, where $\gcd(m, n) = 1$, then the value of $m + n$ equals

(A) 384 (B) 387 (C) 390 (D) 377

Open solution

Q66. JEE Main 2024 (Online) 6th April Evening Shift

Let $\mathrm{P}(\alpha, \beta, \gamma)$ be the image of the point $\mathrm{Q}(3, -3, 1)$ in the line $\frac{x-0}{1} = \frac{y-3}{1} = \frac{z-1}{-1}$ and R be the point $(2, 5, -1)$. If the area of the triangle PQR is λ and $\lambda^2 = 14\,\mathrm{K}$, then K is equal to :

(A) 18 (B) 81 (C) 72 (D) 36

Open solution

Q67. JEE Main 2024 (Online) 6th April Morning Shift

The shortest distance between the lines $\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}$ and $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$ is

(A) $8\sqrt{3}$ (B) $6\sqrt{3}$ (C) $5\sqrt{3}$ (D) $4\sqrt{3}$

Open solution

Q68. JEE Main 2024 (Online) 5th April Evening Shift

Let (α, β, γ) be the image of the point $(8, 5, 7)$ in the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{5}$. Then $\alpha + \beta + \gamma$ is equal to :

(A) 16 (B) 20 (C) 18 (D) 14

Open solution

Q69. JEE Main 2024 (Online) 5th April Morning Shift

If the line $\frac{2-x}{3} = \frac{3y-2}{4\lambda+1} = 4 - z$ makes a right angle with the line $\frac{x+3}{3\mu} = \frac{1-2y}{6} = \frac{5-z}{7}$, then $4\lambda + 9\mu$ is equal to :

(A) 4 (B) 13 (C) 5 (D) 6

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Q70. JEE Main 2024 (Online) 5th April Morning Shift

Let d be the distance of the point of intersection of the lines $\frac{x+6}{3}=\frac{y}{2}=\frac{z+1}{1}$ and $\frac{x-7}{4}=\frac{y-9}{3}=\frac{z-4}{2}$ from the point $(7,8,9)$. Then d^2+6 is equal to :

(A) 75 (B) 78 (C) 72 (D) 69

Open solution

Q71. JEE Main 2024 (Online) 4th April Evening Shift

Let P be the point of intersection of the lines $\frac{x-2}{1}=\frac{y-4}{5}=\frac{z-2}{1}$ and $\frac{x-3}{2}=\frac{y-2}{3}=\frac{z-3}{2}$. Then, the shortest distance of P from the line $4x=2y=z$ is

(A) $\frac{3\sqrt{14}}{7}$ (B) $\frac{5\sqrt{14}}{7}$ (C) $\frac{\sqrt{14}}{7}$ (D) $\frac{6\sqrt{14}}{7}$

Open solution

Q72. JEE Main 2024 (Online) 4th April Morning Shift

Let the point, on the line passing through the points $P(1,-2,3)$ and $Q(5,-4,7)$, farther from the origin and at a distance of 9 units from the point P, be (α,β,γ). Then $\alpha^2+\beta^2+\gamma^2$ is equal to :

(A) 150 (B) 155 (C) 160 (D) 165

Open solution

Q73. JEE Main 2024 (Online) 1st February Evening Shift

Let P and Q be the points on the line $\frac{x+3}{8}=\frac{y-4}{2}=\frac{z+1}{2}$ which are at a distance of 6 units from the point $\mathrm{R}(1,2,3)$. If the centroid of the triangle PQR is (α,β,γ), then $\alpha^2+\beta^2+\gamma^2$ is :

(A) 18 (B) 24 (C) 26 (D) 36

Open solution

Q74. JEE Main 2024 (Online) 1st February Evening Shift

If the mirror image of the point $P(3,4,9)$ in the line

$\frac{x-1}{3}=\frac{y+1}{2}=\frac{z-2}{1}$ is (α,β,γ), then 14 $(\alpha+\beta+\gamma)$ is :

(A) 102 (B) 138 (C) 132 (D) 108

Open solution

Q75. JEE Main 2024 (Online) 1st February Morning Shift

If the shortest distance between the lines

$\frac{x-\lambda}{-2}=\frac{y-2}{1}=\frac{z-1}{1}$ and $\frac{x-\sqrt{3}}{1}=\frac{y-1}{-2}=\frac{z-2}{1}$ is 1 , then the sum of all possible values of λ is :

(A) 0 (B) $2\sqrt{3}$ (C) $3\sqrt{3}$ (D) $-2\sqrt{3}$

Open solution

Q76. JEE Main 2024 (Online) 31st January Evening Shift

Let (α,β,γ) be the mirror image of the point $(2,3,5)$ in the line $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$. Then, $2\alpha+3\beta+4\gamma$ is equal to

(A) 32 (B) 31 (C) 33 (D) 34

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Q77. JEE Main 2024 (Online) 31st January Evening Shift

The shortest distance, between lines L_1 and L_2, where $L_1 : \frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+4}{2}$ and L_2 is the line, passing through the points $\mathrm{A}(-4,4,3), \mathrm{B}(-1,6,3)$ and perpendicular to the line $\frac{x-3}{-2} = \frac{y}{3} = \frac{z-1}{1}$, is

(A) $\frac{141}{\sqrt{221}}$ (B) $\frac{24}{\sqrt{117}}$ (C) $\frac{42}{\sqrt{117}}$ (D) $\frac{121}{\sqrt{221}}$

Open solution

Q78. JEE Main 2024 (Online) 31st January Morning Shift

The distance of the point $Q(0,2,-2)$ from the line passing through the point $P(5,-4,3)$ and perpendicular to the lines $\vec{r} = (-3\hat{i} + 2\hat{k}) + \lambda(2\hat{i} + 3\hat{j} + 5\hat{k}), \lambda \in \mathbb{R}$ and $\vec{r} = (\hat{i} - 2\hat{j} + \hat{k}) + \mu(-\hat{i} + 3\hat{j} + 2\hat{k}), \mu \in \mathbb{R}$ is :

(A) $\sqrt{74}$ (B) $\sqrt{86}$ (C) $\sqrt{54}$ (D) $\sqrt{20}$

Open solution

Q79. JEE Main 2024 (Online) 30th January Evening Shift

Let $L_1 : \vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \lambda(\hat{i} - \hat{j} + 2\hat{k}), \lambda \in \mathbb{R}$,

$L_2 : \vec{r} = (\hat{j} - \hat{k}) + \mu(3\hat{i} + \hat{j} + p\hat{k}), \mu \in \mathbb{R}$, and $L_3 : \vec{r} = \delta(\ell\hat{i} + m\hat{j} + n\hat{k}), \delta \in \mathbb{R}$

be three lines such that L_1 is perpendicular to L_2 and L_3 is perpendicular to both L_1 and L_2. Then, the point which lies on L_3 is

(A) $(1,7,-4)$ (B) $(1,-7,4)$ (C) $(-1,7,4)$ (D) $(-1,-7,4)$

Open solution

Q80. JEE Main 2024 (Online) 30th January Morning Shift

Let (α, β, γ) be the foot of perpendicular from the point $(1,2,3)$ on the line $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3}$. Then $19(\alpha + \beta + \gamma)$ is equal to :

(A) 99 (B) 102 (C) 101 (D) 100

Open solution

Q81. JEE Main 2024 (Online) 29th January Evening Shift

Let $\mathrm{P}(3,2,3), \mathrm{Q}(4,6,2)$ and $\mathrm{R}(7,3,2)$ be the vertices of $\triangle \mathrm{PQR}$. Then, the angle $\angle \mathrm{QPR}$ is

(A) $\cos^{-1}\left(\frac{7}{18}\right)$ (B) $\frac{\pi}{6}$

(C) $\cos^{-1}\left(\frac{1}{18}\right)$ (D) $\frac{\pi}{3}$

Open solution

Q82. JEE Main 2024 (Online) 29th January Morning Shift

Let PQR be a triangle with $R(-1,4,2)$. Suppose $M(2,1,2)$ is the mid point of PQ. The distance of the centroid of $\triangle \mathrm{PQR}$ from the point of intersection of the lines $\frac{x-2}{0} = \frac{y}{2} = \frac{z+3}{-1}$ and $\frac{x-1}{1} = \frac{y+3}{-3} = \frac{z+1}{1}$ is

(A) 69 (B) $\sqrt{99}$ (C) $\sqrt{69}$ (D) 9

Open solution

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Q83. JEE Main 2024 (Online) 27th January Evening Shift

Let the image of the point $(1, 0, 7)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ be the point (α, β, γ). Then which one of the following points lies on the line passing through (α, β, γ) and making angles $\frac{2\pi}{3}$ and $\frac{3\pi}{4}$ with y-axis and z-axis respectively and an acute angle with x-axis ?

(A) $(1, -2, 1+\sqrt{2})$ (B) $(3, -4, 3+2\sqrt{2})$ (C) $(3, 4, 3-2\sqrt{2})$ (D) $(1, 2, 1-\sqrt{2})$

Open solution

Q84. JEE Main 2024 (Online) 27th January Morning Shift

The distance of the point $(7, -2, 11)$ from the line

$\frac{x-6}{1} = \frac{y-4}{0} = \frac{z-8}{3}$ along the line $\frac{x-5}{2} = \frac{y-1}{-3} = \frac{z-5}{6}$, is :

(A) 12 (B) 18 (C) 21 (D) 14

Open solution

Q85. JEE Main 2024 (Online) 27th January Morning Shift

If the shortest distance between the lines

$\frac{x-4}{1} = \frac{y+1}{2} = \frac{z}{-3}$ and $\frac{x-\lambda}{2} = \frac{y+1}{4} = \frac{z-2}{-5}$ is $\frac{6}{\sqrt{5}}$, then the sum of all possible values of λ is :

(A) 10 (B) 5 (C) 7 (D) 8

Open solution

Q86. JEE Main 2023 (Online) 15th April Morning Shift

Let S be the set of all values of λ, for which the shortest distance between the lines

$\frac{x-\lambda}{0} = \frac{y-3}{4} = \frac{z+6}{1}$ and $\frac{x+\lambda}{3} = \frac{y}{-4} = \frac{z-6}{0}$ is 13. Then $8\left|\sum_{\lambda \in S} \lambda\right|$ is equal to :

(A) 306 (B) 304 (C) 308 (D) 302

Open solution

Q87. JEE Main 2023 (Online) 13th April Evening Shift

The line, that is coplanar to the line $\frac{x+3}{-3} = \frac{y-1}{1} = \frac{z-5}{5}$, is :

(A) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{4}$ (B) $\frac{x+1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$

(C) $\frac{x-1}{-1} = \frac{y-2}{2} = \frac{z-5}{5}$ (D) $\frac{x+1}{1} = \frac{y-2}{2} = \frac{z-5}{5}$

Open solution

Q88. JEE Main 2023 (Online) 12th April Morning Shift

Let the lines $l_1 : \frac{x+5}{3} = \frac{y+4}{1} = \frac{z-\alpha}{-2}$ and $l_2 : 3x + 2y + z - 2 = 0 = x - 3y + 2z - 13$ be coplanar. If the point $\mathrm{P}(a, b, c)$ on l_1 is nearest to the point $\mathrm{Q}(-4, -3, 2)$, then $|a| + |b| + |c|$ is equal to

(A) 12 (B) 14 (C) 10 (D) 8

Open solution

Q89. JEE Main 2023 (Online) 10th April Morning Shift

The shortest distance between the lines $\frac{x+2}{1} = \frac{y}{-2} = \frac{z-5}{2}$ and $\frac{x-4}{1} = \frac{y-1}{2} = \frac{z+3}{0}$ is :

(A) 8 (B) 7 (C) 6 (D) 9

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Q90. JEE Main 2023 (Online) 8th April Morning Shift

The shortest distance between the lines $\frac{x-4}{4}=\frac{y+2}{5}=\frac{z+3}{3}$ and $\frac{x-1}{3}=\frac{y-3}{4}=\frac{z-4}{2}$ is :

(A) $3\sqrt{6}$ (B) $6\sqrt{2}$ (C) $6\sqrt{3}$ (D) $2\sqrt{6}$

Open solution

Q91. JEE Main 2023 (Online) 6th April Morning Shift

One vertex of a rectangular parallelopiped is at the origin O and the lengths of its edges along x, y and z axes are $3, 4$ and 5 units respectively. Let P be the vertex $(3, 4, 5)$. Then the shortest distance between the diagonal OP and an edge parallel to z axis, not passing through O or P is :

(A) $\frac{12}{\sqrt{5}}$ (B) $12\sqrt{5}$ (C) $\frac{12}{5}$ (D) $\frac{12}{5\sqrt{5}}$

Open solution

Q92. JEE Main 2023 (Online) 1st February Morning Shift

The shortest distance between the lines

$\frac{x-5}{1}=\frac{y-2}{2}=\frac{z-4}{-3}$ and

$\frac{x+3}{1}=\frac{y+5}{4}=\frac{z-1}{-5}$ is :

(A) $7\sqrt{3}$ (B) $5\sqrt{3}$ (C) $4\sqrt{3}$ (D) $6\sqrt{3}$

Open solution

Q93. JEE Main 2023 (Online) 31st January Morning Shift

Let the shortest distance between the lines

$L: \frac{x-5}{-2}=\frac{y-\lambda}{0}=\frac{z+\lambda}{1}, \lambda \geq 0$ and

$L_1: x+1=y-1=4-z$ be $2\sqrt{6}$. If (α, β, γ) lies on L,

then which of the following is NOT possible?

(A) $\alpha+2\gamma=24$ (B) $2\alpha+\gamma=7$ (C) $\alpha-2\gamma=19$ (D) $2\alpha-\gamma=9$

Open solution

Q94. JEE Main 2023 (Online) 29th January Evening Shift

The shortest distance between the lines $\frac{x-1}{2}=\frac{y+8}{-7}=\frac{z-4}{5}$ and $\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-6}{-3}$ is :

(A) $2\sqrt{3}$ (B) $3\sqrt{3}$ (C) $4\sqrt{3}$ (D) $5\sqrt{3}$

Open solution

Q95. JEE Main 2023 (Online) 25th January Evening Shift

The foot of perpendicular of the point (2, 0, 5) on the line $\frac{x+1}{2}=\frac{y-1}{5}=\frac{z+1}{-1}$ is (α, β, γ). Then, which of the following is NOT correct?

(A) $\frac{\alpha}{\beta}=-8$ (B) $\frac{\alpha\beta}{\gamma}=\frac{4}{15}$

(C) $\frac{\beta}{\gamma}=-5$ (D) $\frac{\gamma}{\alpha}=\frac{5}{8}$

Open solution

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Q96. JEE Main 2023 (Online) 25th January Evening Shift

The shortest distance between the lines $x+1=2y=-12z$ and $x=y+2=6z-6$ is :

(A) 3 (B) $\frac{5}{2}$ (C) $\frac{3}{2}$ (D) 2

Open solution

Q97. JEE Main 2023 (Online) 25th January Morning Shift

The distance of the point P(4, 6, -2) from the line passing through the point $(-3, 2, 3)$ and parallel to a line with direction ratios 3, 3, -1 is equal to :

(A) 3 (B) $\sqrt{14}$ (C) $\sqrt{6}$ (D) $2\sqrt{3}$

Open solution

Q98. JEE Main 2023 (Online) 25th January Morning Shift

Consider the lines L_1 and L_2 given by

$L_1 : \frac{x-1}{2}=\frac{y-3}{1}=\frac{z-2}{2}$

$L_2 : \frac{x-2}{1}=\frac{y-2}{2}=\frac{z-3}{3}$.

A line L_3 having direction ratios 1, -1, -2, intersects L_1 and L_2 at the points P and Q respectively. Then the length of line segment PQ is

(A) $4\sqrt{3}$ (B) $2\sqrt{6}$ (C) 4 (D) $3\sqrt{2}$

Open solution

Q99. JEE Main 2022 (Online) 27th July Evening Shift

If the length of the perpendicular drawn from the point $P(a, 4, 2)$, a > 0 on the line $\frac{x+1}{2}=\frac{y-3}{3}=\frac{z-1}{-1}$ is $2\sqrt{6}$ units and $Q(\alpha_1, \alpha_2, \alpha_3)$ is the image of the point P in this line, then $a+\sum_{i=1}^{3}\alpha_i$ is equal to :

(A) 7 (B) 8 (C) 12 (D) 14

Open solution

Q100. JEE Main 2022 (Online) 26th July Morning Shift

The length of the perpendicular from the point $(1, -2, 5)$ on the line passing through $(1, 2, 4)$ and parallel to the line $x+y-z=0=x-2y+3z-5$ is :

(A) $\sqrt{\frac{21}{2}}$ (B) $\sqrt{\frac{9}{2}}$ (C) $\sqrt{\frac{73}{2}}$ (D) 1

Open solution

Q101. JEE Main 2022 (Online) 25th July Evening Shift

The shortest distance between the lines $\frac{x+7}{-6}=\frac{y-6}{7}=z$ and $\frac{7-x}{2}=y-2=z-6$ is :

(A) $2\sqrt{29}$ (B) 1 (C) $\sqrt{\frac{37}{29}}$ (D) $\frac{\sqrt{29}}{2}$

Open solution

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Q102. JEE Main 2022 (Online) 27th June Evening Shift

The shortest distance between the lines

$\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-1}{-1}$ and $\frac{x+3}{2} = \frac{y-6}{1} = \frac{z-5}{3}$, is :

(A) $\frac{18}{\sqrt{5}}$ (B) $\frac{22}{3\sqrt{5}}$ (C) $\frac{46}{3\sqrt{5}}$ (D) $6\sqrt{3}$

Open solution

Q103. JEE Main 2022 (Online) 27th June Morning Shift

If two straight lines whose direction cosines are given by the relations $l + m - n = 0$, $3l^2 + m^2 + cnl = 0$ are parallel, then the positive value of c is :

(A) 6 (B) 4 (C) 3 (D) 2

Open solution

Q104. JEE Main 2022 (Online) 26th June Morning Shift

If the two lines $l_1 : \frac{x-2}{3} = \frac{y+1}{-2}$, $z = 2$ and $l_2 : \frac{x-1}{1} = \frac{2y+3}{\alpha} = \frac{z+5}{2}$ are perpendicular, then an angle between the lines l_2 and $l_3 : \frac{1-x}{3} = \frac{2y-1}{-4} = \frac{z}{4}$ is :

(A) $\cos^{-1}\left(\frac{29}{4}\right)$ (B) $\sec^{-1}\left(\frac{29}{4}\right)$

(C) $\cos^{-1}\left(\frac{2}{29}\right)$ (D) $\cos^{-1}\left(\frac{2}{\sqrt{29}}\right)$

Open solution

Q105. JEE Main 2022 (Online) 24th June Evening Shift

If the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{\lambda}$ and $\frac{x-2}{1} = \frac{y-4}{4} = \frac{z-5}{5}$ is $\frac{1}{\sqrt{3}}$, then the sum of all possible value of λ is :

(A) 16 (B) 6 (C) 12 (D) 15

Open solution

Q106. JEE Main 2021 (Online) 1st September Evening Shift

The distance of line $3y - 2z - 1 = 0 = 3x - z + 4$ from the point (2, −1, 6) is :

(A) $\sqrt{26}$ (B) $2\sqrt{5}$ (C) $2\sqrt{6}$ (D) $4\sqrt{2}$

Open solution

Q107. JEE Main 2021 (Online) 27th August Evening Shift

The angle between the straight lines, whose direction cosines are given by the equations 2l + 2m − n = 0 and mn + nl + lm = 0, is :

(A) $\frac{\pi}{2}$ (B) $\pi - \cos^{-1}\left(\frac{4}{9}\right)$

(C) $\cos^{-1}\left(\frac{8}{9}\right)$ (D) $\frac{\pi}{3}$

Open solution

Q108. JEE Main 2021 (Online) 22th July Evening Shift

If the shortest distance between the straight lines $3(x-1) = 6(y-2) = 2(z-1)$ and $4(x-2) = 2(y-\lambda) = (z-3), \lambda \in R$ is $\frac{1}{\sqrt{38}}$, then the integral value of λ is equal to :

(A) 3 (B) 2 (C) 5 (D) −1

Open solution

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Q109. JEE Main 2021 (Online) 20th July Evening Shift

The lines x = ay − 1 = z − 2 and x = 3y − 2 = bz − 2, (ab ≠ 0) are coplanar, if :

(A) b = 1, a∈R − {0}

(B) a = 1, b∈R − {0}

(C) a = 2, b = 2

(D) a = 2, b = 3

Open solution

Q110. JEE Main 2021 (Online) 16th March Evening Shift

If the foot of the perpendicular from point (4, 3, 8) on the line $L_1 : \frac{x-a}{l} = \frac{y-2}{3} = \frac{z-b}{4}$, $l \neq 0$ is (3, 5, 7), then the shortest distance between the line L_1 and line $L_2 : \frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$ is equal to :

(A) $\frac{1}{\sqrt{6}}$

(B) $\frac{1}{2}$

(C) $\frac{1}{\sqrt{3}}$

(D) $\sqrt{\frac{2}{3}}$

Open solution

Q111. JEE Main 2021 (Online) 16th March Morning Shift

Let the position vectors of two points P and Q be $3\hat{i} - \hat{j} + 2\hat{k}$ and $\hat{i} + 2\hat{j} - 4\hat{k}$, respectively. Let R and S be two points such that the direction ratios of lines PR and QS are (4, −1, 2) and (−2, 1, −2), respectively. Let lines PR and QS intersect at T. If the vector $\overrightarrow{TA}$ is perpendicular to both $\overrightarrow{PR}$ and $\overrightarrow{QS}$ and the length of vector $\overrightarrow{TA}$ is $\sqrt{5}$ units, then the modulus of a position vector of A is :

(A) $\sqrt{171}$

(B) $\sqrt{227}$

(C) $\sqrt{482}$

(D) $\sqrt{5}$

Open solution

Q112. JEE Main 2021 (Online) 25th February Morning Shift

The equation of the line through the point (0, 1, 2) and perpendicular to the line

$\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{-2}$ is :

(A) $\frac{x}{3} = \frac{y-1}{-4} = \frac{z-2}{3}$

(B) $\frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{-3}$

(C) $\frac{x}{-3} = \frac{y-1}{4} = \frac{z-2}{3}$

(D) $\frac{x}{3} = \frac{y-1}{4} = \frac{z-2}{3}$

Open solution

Q113. JEE Main 2021 (Online) 25th February Morning Shift

Let α be the angle between the lines whose direction cosines satisfy the equations l + m − n = 0 and $l^2 + m^2 - n^2 = 0$. Then the value of $\sin^4\alpha + \cos^4\alpha$ is :

(A) $\frac{3}{8}$

(B) $\frac{3}{4}$

(C) $\frac{1}{2}$

(D) $\frac{5}{8}$

Open solution

Q114. JEE Main 2021 (Online) 24th February Evening Shift

Let a, b∈R. If the mirror image of the point P(a, 6, 9) with respect to the line

$\frac{x-3}{7} = \frac{y-2}{5} = \frac{z-1}{-9}$ is (20, b, −a−9), then | a + b |, is equal to :

(A) 88

(B) 90

(C) 86

(D) 84

Open solution

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Q115. JEE Main 2020 (Online) 5th September Morning Slot

If (a, b, c) is the image of the point (1, 2, -3) in the line $\frac{x+1}{2} = \frac{y-3}{-2} = \frac{z}{-1}$, then a + b + c is :

Open solution

(A) 1 (B) 2 (C) 3 (D) -1

Q116. JEE Main 2020 (Online) 3rd September Morning Slot

The lines
$\overrightarrow{r} = \left(\hat{i} - \hat{j}\right) + l\left(2\hat{i} + \hat{k}\right)$ and
$\overrightarrow{r} = \left(2\hat{i} - \hat{j}\right) + m\left(\hat{i} + \hat{j} + \hat{k}\right)$

Open solution

(A) do not intersect for any values of l and m

(B) intersect for all values of l and m

(C) intersect when l = 2 and m = $\frac{1}{2}$

(D) intersect when l = 1 and m = 2

Q117. JEE Main 2020 (Online) 8th January Morning Slot

The shortest distance between the lines

$\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}$ and

$\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$ is :

Open solution

(A) 3 (B) $\frac{7}{2}\sqrt{30}$ (C) $3\sqrt{30}$ (D) $2\sqrt{30}$

Q118. JEE Main 2019 (Online) 10th April Evening Slot

The distance of the point having position vector $-\hat{i} + 2\hat{j} + 6\hat{k}$ from the straight line passing through the point (2, 3, – 4) and parallel to the vector, $6\hat{i} + 3\hat{j} - 4\hat{k}$ is :

Open solution

(A) 6 (B) 7 (C) $2\sqrt{13}$ (D) $4\sqrt{3}$

Q119. JEE Main 2019 (Online) 10th April Morning Slot

If the length of the perpendicular from the point (β, 0, β) ($\beta \neq 0$) to the line, $\frac{x}{1} = \frac{y-1}{0} = \frac{z+1}{-1}$ is $\sqrt{\frac{3}{2}}$, then β is equal to :

Open solution

(A) 2 (B) 1 (C) -2 (D) -1

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Q120. JEE Main 2019 (Online) 9th April Evening Slot

The vertices B and C of a ΔABC lie on the line,

$\frac{x+2}{3} = \frac{y-1}{0} = \frac{z}{4}$ such that BC = 5 units.

Then the area (in sq. units) of this triangle, given that the point A(1, −1, 2), is :

(A) 6 (B) $5\sqrt{17}$ (C) $\sqrt{34}$ (D) $2\sqrt{34}$

Open solution

Q121. JEE Main 2019 (Online) 8th April Evening Slot

If a point R(4, y, z) lies on the line segment joining the points P(2, −3, 4) and Q(8, 0, 10), then the distance of R from the origin is :

(A) $2\sqrt{14}$ (B) $\sqrt{53}$ (C) $2\sqrt{21}$ (D) 6

Open solution

Q122. JEE Main 2019 (Online) 8th April Morning Slot

The length of the perpendicular from the point (2, −1, 4) on the straight line,

$\frac{x+3}{10} = \frac{y-2}{-7} = \frac{z}{1}$ is :

(A) less than 2 (B) greater than 4

(C) greater than 2 but less than 3 (D) greater than 3 but less than 4

Open solution

Q123. JEE Main 2019 (Online) 9th January Evening Slot

If the lines x = ay + b, z = cy + d and x = a'z + b', y = c'z + d' are perpendicular, then :

(A) ab' + bc' + 1 = 0 (B) cc' + a + a' = 0

(C) bb' + cc' + 1 = 0 (D) aa' + c + c' = 0

Open solution

Q124. JEE Main 2018 (Online) 16th April Morning Slot

If the angle between the lines, $\frac{x}{2} = \frac{y}{2} = \frac{z}{1}$

and $\frac{5-x}{-2} = \frac{7y-14}{p} = \frac{z-3}{4}$ is $\cos^{-1}\left(\frac{2}{3}\right)$, then p is equal to :

(A) $\frac{7}{2}$ (B) $\frac{2}{7}$ (C) $-\frac{7}{4}$ (D) $-\frac{4}{7}$

Open solution

Q125. JEE Main 2018 (Online) 15th April Evening Slot

An angle between the lines whose direction cosines are given by the equations,
l + 3m + 5n = 0 and 5lm − 2mn + 6nl = 0, is :

(A) $\cos^{-1}\left(\frac{1}{3}\right)$ (B) $\cos^{-1}\left(\frac{1}{4}\right)$

(C) $\cos^{-1}\left(\frac{1}{6}\right)$ (D) $\cos^{-1}\left(\frac{1}{8}\right)$

Open solution

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Q126. JEE Main 2016 (Online) 10th April Morning Slot

ABC is a triangle in a plane with vertices

A(2, 3, 5), B(−1, 3, 2) and C(λ, 5, μ).

If the median through A is equally inclined to the coordinate axes, then the value of ($\lambda^3 + \mu^3 + 5$) is :

(A) 1130 (B) 1348 (C) 676 (D) 1077

Open solution

Q127. JEE Main 2016 (Online) 9th April Morning Slot

The shortest distance between the lines $\frac{x}{2} = \frac{y}{2} = \frac{z}{1}$ and $\frac{x+2}{-1} = \frac{y-4}{8} = \frac{z-5}{4}$ lies in the interval :

(A) [0, 1) (B) [1, 2) (C) (2, 3] (D) (3, 4]

Open solution

Q128. JEE Main 2015 (Online) 11th April Morning Slot

The shortest distance between the z-axis and the line

x + y + 2z − 3 = 0 = 2x + 3y + 4z − 4, is :

(A) 1 (B) 2 (C) 3 (D) 4

Open solution

Q129. JEE Main 2015 (Online) 10th April Morning Slot

If the shortest distance between the lines

$\frac{x-1}{\alpha} = \frac{y+1}{-1} = \frac{z}{1}$, ($\alpha \neq -1$) and

x + y + z + 1 = 0

= 2x − y + z + 3 is $\frac{1}{\sqrt{3}}$, then a value of α is :

(A) $-\frac{16}{19}$ (B) $-\frac{19}{16}$ (C) $\frac{32}{19}$ (D) $\frac{19}{32}$

Open solution

Q130. JEE Main 2014 (Online) 19th April Morning Slot

Equation of the line of the shortest distance between the lines $\frac{x}{1} = \frac{y}{-1} = \frac{z}{1}$ and $\frac{x-1}{0} = \frac{y+1}{-2} = \frac{z}{1}$ is :

(A) $\frac{x}{1} = \frac{y}{-1} = \frac{z}{-2}$ (B) $\frac{x-1}{1} = \frac{y+1}{-1} = \frac{z}{-2}$

(C) $\frac{x-1}{1} = \frac{y+1}{-1} = \frac{z}{1}$ (D) $\frac{x}{-2} = \frac{y}{1} = \frac{z}{2}$

Open solution

Q131. JEE Main 2014 (Online) 11th April Morning Slot

Let A (2, 3, 5), B(−1, 3, 2) and C(λ, 5, μ) be the vertices of a ΔABC. If the median through A is equally inclined to the coordinate axes, then :

(A) $5\lambda - 8\mu = 0$ (B) $8\lambda - 5\mu = 0$ (C) $10\lambda - 7\mu = 0$ (D) $7\lambda - 10\mu = 0$

Open solution

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Q132. JEE Main 2014 (Online) 9th April Morning Slot

A line in the 3-dimensional space makes an angle $\theta(0 < \theta \leq \frac{\pi}{2})$ with both the x and y axes. Then the set of all values of θ is the interval :

(A) $\left(0, \frac{\pi}{4}\right]$ (B) $\left[\frac{\pi}{6}, \frac{\pi}{3}\right]$

(C) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$ (D) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$

Open solution

Q133. JEE Main 2014 (Offline)

The angle between the lines whose direction cosines satisfy the equations $l + m + n = 0$ and $l^2 = m^2 + n^2$ is :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{4}$

Open solution

Q134. JEE Main 2013 (Online) 25th April Morning Slot

Let ABC be a triangle with vertices at points A (2, 3, 5), B $(-1, 3, 2)$ and C$(\lambda, 5, \mu)$ in three dimensional space. If the median through A is equally inclined with the axes, then (λ, μ) is equal to :

(A) (10, 7) (B) (7, 5) (C) (7, 10) (D) (5, 7)

Open solution

Q135. JEE Main 2013 (Online) 23th April Morning Slot

If the projections of a line segment on the x, y and z-axes in 3-dimensional space are 2, 3 and 6 respectively, then the length of the line segment is :

(A) 12 (B) 7 (C) 9 (D) 6

Open solution

Q136. JEE Main 2013 (Online) 23th April Morning Slot

If two lines L_1 and L_2 in space, are defined by

$L_1 = \{x = \sqrt{\lambda}y + (\sqrt{\lambda} - 1), z = (\sqrt{\lambda} - 1)\, y + \sqrt{\lambda}\}$ and

$L_2 = \{x = \sqrt{\mu}y + (1 - \sqrt{\mu}), z = (1 - \sqrt{\mu})\, y + \sqrt{\mu}\}$

then L_1 is perpendicular to L_2, for all non-negative reals λ and μ, such that :

(A) $\sqrt{\lambda} + \sqrt{\mu} = 1$ (B) $\lambda \neq \mu$

(C) $\lambda + \mu = 0$ (D) $\lambda = \mu$

Open solution

Q137. JEE Main 2013 (Online) 22th April Morning Slot

The acute angle between two lines such that the direction cosines l, m, n, of each of them satisfy the equations $l + m + n = 0$ and $l^2 + m^2 - n^2 = 0$ is :

(A) 15° (B) 30° (C) 60° (D) 45°

Open solution

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Q138. JEE Main 2013 (Online) 9th April Morning Slot

If the lines $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z+1}{3}$ and $\frac{x+2}{2} = \frac{y-k}{3} = \frac{z}{4}$ are coplanar, then the value of k is :

(A) $\frac{11}{2}$ (B) $-\frac{11}{2}$ (C) $\frac{9}{2}$ (D) $-\frac{9}{2}$

Open solution

Q139. JEE Main 2013 (Offline)

If the lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar, then k can have :

(A) any value (B) exactly one value

(C) exactly two values (D) exactly three values

Open solution

Q140. AIEEE 2012 (Online) 26th May

The distance of the point $-\hat{i} + 2\hat{j} + 6\hat{k}$ from the straight line that passes through the point $2\hat{i} + 3\hat{j} - 4\hat{k}$ and is parallel to the vector $6\hat{i} + 3\hat{j} - 4\hat{k}$ is

(A) 9 (B) 8 (C) 7 (D) 10

Open solution

Q141. AIEEE 2012 (Online) 19th May

Statement 1: The shortest distance between the lines $\frac{x}{2} = \frac{y}{-1} = \frac{z}{2}$ and $\frac{x-1}{4} = \frac{y-1}{-2} = \frac{z-1}{4}$ is $\sqrt{2}$.
Statement 2: The shortest distance between two parallel lines is the perpendicular distance from any point on one of the lines to the other line.

(A) Statement 1 is true, Statement 2 is false.

(B) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation for Statement 1.

(C) Statement 1 is false, Statement 2 is true.

(D) Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation for Statement 1.

Open solution

Q142. AIEEE 2012 (Online) 12th May

The coordinates of the foot perpendicular from the point $(1, 0, 0)$ to the line $\frac{x-1}{2} = \frac{y+1}{-3} = \frac{z+10}{8}$ are

(A) $(2, -3, 8)$ (B) $(1, -1, -10)$ (C) $(5, -8, -4)$ (D) $(3, -4, -2)$

Open solution

Q143. AIEEE 2012

If the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect, then k is equal to :

(A) -1 (B) $\frac{2}{9}$ (C) $\frac{9}{2}$ (D) 0

Open solution

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Q144. AIEEE 2011

Statement - 1 : The point $A(1,0,7)$ is the mirror image of the point

$B(1,6,3)$ in the line : $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$

Statement - 2 : The line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$ bisects the line

segment joining $A(1,0,7)$ and $B(1,6,3)$

(A) Statement -1 is true, Statement -2 is true; Statement -2 is **not** a correct explanation for Statement -1.

(B) Statement -1 is true, Statement - 2 is false.

(C) Statement - 1 is false , Statement -2 is true.

(D) Statement -1 is true, Statement -2 is true; Statement -2 is a correct explanation for Statement -1.

Open solution

Q145. AIEEE 2010

A line AB in three-dimensional space makes angles 45° and 120° with the positive x-axis and the positive y-axis respectively. If AB makes an acute angle θ with the positive z-axis, then θ equals :

(A) 45° (B) 60° (C) 75° (D) 30°

Open solution

Q146. AIEEE 2009

The projections of a vector on the three coordinate axis are $6,-3,2$ respectively. The direction cosines of the vector are :

(A) $\frac{6}{5},\frac{-3}{5},\frac{2}{5}$ (B) $\frac{6}{7},\frac{-3}{7},\frac{2}{7}$

(C) $\frac{-6}{7},\frac{-3}{7},\frac{2}{7}$ (D) $6,-3,2$

Open solution

Q147. AIEEE 2008

The line passing through the points $(5,1,a)$ and $(3,b,1)$ crosses the yz-plane at the point $\left(0,\frac{17}{2},\frac{-13}{2}\right)$. Then

(A) $a=2,b=8$ (B) $a=4,b=6$ (C) $a=6,b=4$ (D) $a=8,b=2$

Open solution

Q148. AIEEE 2008

If the straight lines $\frac{x-1}{k}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x-2}{3}=\frac{y-3}{k}=\frac{z-1}{2}$ intersects at a point, then the integer k is equal to

(A) -5 (B) 5 (C) 2 (D) -2

Open solution

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Q149. AIEEE 2007

If a line makes an angle of $\pi/4$ with the positive directions of each of x-axis and y-axis, then the angle that the line makes with the positive direction of the z-axis is :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{3}$

Open solution

Q150. AIEEE 2007

Let L be the line of intersection of the planes $2x+3y+z=1$ and $x+3y+2z=2$. If L makes an angle α with the positive x-axis, then cos α equals

(A) 1 (B) $\frac{1}{\sqrt{2}}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{2}$

Open solution

Q151. AIEEE 2006

The two lines $x=ay+b, z=cy+d$; and $x=a'y+b', z=c'y+d'$ are perpendicular to each other if :

(A) $aa'+cc'=-1$ (B) $aa'+cc'=1$

(C) $\frac{a}{a'}+\frac{c}{c'}=-1$ (D) $\frac{a}{a'}+\frac{c}{c'}=1$

Open solution

Q152. AIEEE 2005

The angle between the lines $2x=3y=-z$ and $6x=-y=-4z$ is :

(A) 0° (B) 90° (C) 45° (D) 30°

Open solution

Q153. AIEEE 2004

A line makes the same angle θ, with each of the x and z axes.

If the angle β, which it makes with y-axis, is such that $\sin^2\beta = 3\sin^2\theta$, then $\cos^2\theta$ equals :

(A) $\frac{2}{5}$ (B) $\frac{1}{5}$ (C) $\frac{3}{5}$ (D) $\frac{2}{3}$

Open solution

Q154. AIEEE 2004

A line with direction cosines proportional to $2, 1, 2$ meets each of the lines $x=y+a=z$ and $x+a=2y=2z$. The co-ordinates of each of the points of intersection are given by :

(A) $(2a, 3a, 3a), (2a, a, a)$ (B) $(3a, 2a, 3a), (a, a, a)$

(C) $(3a, 2a, 3a), (a, a, 2a)$ (D) $(3a, 3a, 3a), (a, a, a)$

Open solution

Q155. AIEEE 2004

If the straight lines
$x=1+s, y=-3-\lambda s, z=1+\lambda s$ and $x=\frac{t}{2}, y=1+t, z=2-t$, with parameters s and t respectively, are co-planar, then λ equals :

(A) 0 (B) -1 (C) $-\frac{1}{2}$ (D) -2

Open solution

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Q156. AIEEE 2003

The two lines $x = ay + b, z = cy + d$ and $x = a'y + b', z = c'y + d'$ will be perpendicular, if and only if :

(A) $aa' + cc' + 1 = 0$

(B) $aa' + bb'cc' + 1 = 0$

(C) $aa' + bb'cc' = 0$

(D) $(a + a')(b + b') + (c + c') = 0$

Open solution

Q157. AIEEE 2003

The lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{z-4}{-k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{1}$ are coplanar if :

(A) $k = 3$ or -2

(B) $k = 0$ or -1

(C) $k = 1$ or -1

(D) $k = 0$ or -3

Open solution

Numerical 42 questions

Q158. JEE Main 2026 (Online) 8th April Evening Shift

Let a line L_1 pass through the origin and be perpendicular to the lines

$\mathrm{L}_2 : \vec{\mathrm{r}} = (3 + \mathrm{t})\hat{i} + (2\mathrm{t} - 1)\hat{j} + (2\mathrm{t} + 4)\hat{k}$ and

$\mathrm{L}_3 : \vec{\mathrm{r}} = (3 + 2\,\mathrm{s})\hat{i} + (3 + 2\,\mathrm{s})\hat{j} + (2 + \mathrm{s})\hat{k}, \mathrm{t}, \mathrm{s} \in \mathbf{R}$.

If $(a, b, c), a \in \mathbf{Z}$, is the point on L_3 at a distance of $\sqrt{17}$ from the point of intersection of L_1 and L_2, then $(\mathrm{a} + \mathrm{b} + \mathrm{c})^2$ is equal to ____ .

Open solution

Q159. JEE Main 2026 (Online) 6th April Evening Shift

Let the image of the point $\mathrm{P}(0, -5, 0)$ in the line $\frac{x-1}{2} = \frac{y}{1} = \frac{z+1}{-2}$ be the point R and the image of the point $\mathrm{Q}\left(0, \frac{-1}{2}, 0\right)$ in the line $\frac{x-1}{-1} = \frac{y+9}{4} = \frac{z+1}{1}$ be the point S . Then the square of the area of the parallelogram PQRS is ____ .

Open solution

Q160. JEE Main 2026 (Online) 28th January Evening Shift

If the distance of the point $P(43, \alpha, \beta)$, $\beta < 0$, from the line $\vec{r} = 4\hat{i} - \hat{k} + \mu(2\hat{i} + 3\hat{k}), \mu \in \mathbb{R}$ along a line with direction ratios $3, -1, 0$ is $13\sqrt{10}$, then $\alpha^2 + \beta^2$ is equal to _____

Open solution

Q161. JEE Main 2026 (Online) 24th January Morning Shift

Let a line L passing through the point $\mathrm{P}(1, 1, 1)$ be perpendicular to the lines $\frac{x-4}{4} = \frac{y-1}{1} = \frac{z-1}{1}$ and $\frac{x-17}{1} = \frac{y-71}{1} = \frac{z}{0}$. Let the line L intersect the $yz-$ plane at the point Q . Another line parallel to L and passing through the point $\mathrm{S}(1, 0, -1)$ intersects the yz-plane at the point R . Then the square of the area of the parallelogram PQRS is equal to ____ .

Open solution

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Q162. JEE Main 2026 (Online) 23rd January Evening Shift

If the image of the point $\mathrm{P}(a, 2, a)$ in the line $\frac{x}{2}=\frac{y+a}{1}=\frac{z}{1}$ is Q and the image of Q in the line $\frac{x-2b}{2}=\frac{y-a}{1}=\frac{z+2b}{-5}$ is P , then $a+b$ is equal to ____ .

Open solution

Q163. JEE Main 2025 (Online) 8th April Evening Shift

Let the area of the triangle formed by the lines $x+2=y-1=z$, $\frac{x-3}{5}=\frac{y}{-1}=\frac{z-1}{1}$ and $\frac{x}{-3}=\frac{y-3}{3}=\frac{z-2}{1}$ be A. Then A^2 is equal to _____.

Open solution

Q164. JEE Main 2025 (Online) 24th January Evening Shift

Let P be the image of the point $\mathrm{Q}(7,-2,5)$ in the line $\mathrm{L}: \frac{x-1}{2}=\frac{y+1}{3}=\frac{z}{4}$ and $\mathrm{R}(5, \mathrm{p}, \mathrm{q})$ be a point on L. Then the square of the area of $\triangle PQR$ is _____.

Open solution

Q165. JEE Main 2025 (Online) 22nd January Morning Shift

Let $\mathrm{L}_1: \frac{x-1}{3}=\frac{y-1}{-1}=\frac{z+1}{0}$ and $\mathrm{L}_2: \frac{x-2}{2}=\frac{y}{0}=\frac{z+4}{\alpha}, \alpha \in \mathbf{R}$, be two lines, which intersect at the point B. If P is the foot of perpendicular from the point $A(1,1,-1)$ on L_2, then the value of $26 \alpha(\mathrm{PB})^2$ is _____ .

Open solution

Q166. JEE Main 2024 (Online) 9th April Evening Shift

The square of the distance of the image of the point $(6,1,5)$ in the line $\frac{x-1}{3}=\frac{y}{2}=\frac{z-2}{4}$, from the origin is _____.

Open solution

Q167. JEE Main 2024 (Online) 8th April Evening Shift

Let $\mathrm{P}(\alpha, \beta, \gamma)$ be the image of the point $\mathrm{Q}(1,6,4)$ in the line $\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$. Then $2 \alpha+\beta+\gamma$ is equal to _____

Open solution

Q168. JEE Main 2024 (Online) 6th April Evening Shift

If the shortest distance between the lines $\frac{x-\lambda}{3}=\frac{y-2}{-1}=\frac{z-1}{1}$ and $\frac{x+2}{-3}=\frac{y+5}{2}=\frac{z-4}{4}$ is $\frac{44}{\sqrt{30}}$, then the largest possible value of $|\lambda|$ is equal to _____.

Open solution

Q169. JEE Main 2024 (Online) 6th April Morning Shift

Let P be the point $(10,-2,-1)$ and Q be the foot of the perpendicular drawn from the point $R(1,7,6)$ on the line passing through the points $(2,-5,11)$ and $(-6,7,-5)$. Then the length of the line segment PQ is equal to _____.

Open solution

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Q170. JEE Main 2024 (Online) 5th April Evening Shift

Let the point $(-1, \alpha, \beta)$ lie on the line of the shortest distance between the lines $\frac{x+2}{-3} = \frac{y-2}{4} = \frac{z-5}{2}$ and $\frac{x+2}{-1} = \frac{y+6}{2} = \frac{z-1}{0}$. Then $(\alpha - \beta)^2$ is equal to _____.

Open solution

Q171. JEE Main 2024 (Online) 4th April Evening Shift

Consider a line L passing through the points $\mathrm{P}(1, 2, 1)$ and $\mathrm{Q}(2, 1, -1)$. If the mirror image of the point $\mathrm{A}(2, 2, 2)$ in the line L is (α, β, γ), then $\alpha + \beta + 6\gamma$ is equal to _____.

Open solution

Q172. JEE Main 2024 (Online) 1st February Morning Shift

Let the line of the shortest distance between the lines

$\mathrm{L}_1 : \vec{\mathrm{r}} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and

$\mathrm{L}_2 : \vec{\mathrm{r}} = (4\hat{i} + 5\hat{j} + 6\hat{k}) + \mu(\hat{i} + \hat{j} - \hat{k})$

intersect L_1 and L_2 at P and Q respectively. If (α, β, γ) is the mid point of the line segment PQ, then $2(\alpha + \beta + \gamma)$ is equal to _____.

Open solution

Q173. JEE Main 2024 (Online) 31st January Evening Shift

A line passes through $A(4, -6, -2)$ and $B(16, -2, 4)$. The point $P(a, b, c)$, where a, b, c are non-negative integers, on the line AB lies at a distance of 21 units, from the point A. The distance between the points $P(a, b, c)$ and $Q(4, -12, 3)$ is equal to _____.

Open solution

Q174. JEE Main 2024 (Online) 31st January Morning Shift

Let Q and R be the feet of perpendiculars from the point $\mathrm{P}(a, a, a)$ on the lines $x = y, z = 1$ and $x = -y, z = -1$ respectively. If $\angle\mathrm{QPR}$ is a right angle, then $12a^2$ is equal to _____.

Open solution

Q175. JEE Main 2024 (Online) 30th January Evening Shift

Let a line passing through the point $(-1, 2, 3)$ intersect the lines $L_1 : \frac{x-1}{3} = \frac{y-2}{2} = \frac{z+1}{-2}$ at $M(\alpha, \beta, \gamma)$ and $L_2 : \frac{x+2}{-3} = \frac{y-2}{-2} = \frac{z-1}{4}$ at $N(a, b, c)$. Then, the value of $\frac{(\alpha+\beta+\gamma)^2}{(a+b+c)^2}$ equals _____.

Open solution

Q176. JEE Main 2024 (Online) 30th January Morning Shift

If d_1 is the shortest distance between the lines $x + 1 = 2y = -12z, x = y + 2 = 6z - 6$ and d_2 is the shortest distance between the lines $\frac{x-1}{2} = \frac{y+8}{-7} = \frac{z-4}{5}, \frac{x-1}{2} = \frac{y-2}{1} = \frac{z-6}{-3}$, then the value of $\frac{32\sqrt{3}\,\mathrm{d}_1}{\mathrm{d}_2}$ is :

Open solution

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Q177. JEE Main 2024 (Online) 29th January Evening Shift

Let O be the origin, and M and N be the points on the lines $\frac{x-5}{4}=\frac{y-4}{1}=\frac{z-5}{3}$ and $\frac{x+8}{12}=\frac{y+2}{5}=\frac{z+11}{9}$ respectively such that MN is the shortest distance between the given lines. Then $\overrightarrow{OM}\cdot\overrightarrow{ON}$ is equal to _____.

Open solution

Q178. JEE Main 2024 (Online) 29th January Morning Shift

A line with direction ratios $2,1,2$ meets the lines $x=y+2=z$ and $x+2=2y=2z$ respectively at the points P and Q. If the length of the perpendicular from the point $(1,2,12)$ to the line PQ is l, then l^2 is _____.

Open solution

Q179. JEE Main 2024 (Online) 27th January Evening Shift

The lines $\frac{x-2}{2}=\frac{y}{-2}=\frac{z-7}{16}$ and $\frac{x+3}{4}=\frac{y+2}{3}=\frac{z+2}{1}$ intersect at the point P. If the distance of P from the line $\frac{x+1}{2}=\frac{y-1}{3}=\frac{z-1}{1}$ is l, then $14l^2$ is equal to _____.

Open solution

Q180. JEE Main 2023 (Online) 11th April Morning Shift

Let a line l pass through the origin and be perpendicular to the lines

$l_1:\vec{r}=(\hat{i}-11\hat{j}-7\hat{k})+\lambda(\hat{i}+2\hat{j}+3\hat{k}),\lambda\in\mathbb{R}$ and

$l_2:\vec{r}=(-\hat{i}+\hat{k})+\mu(2\hat{i}+2\hat{j}+\hat{k}),\mu\in\mathbb{R}$.

If P is the point of intersection of l and l_1, and $Q(\alpha,\beta,\gamma)$ is the foot of perpendicular from P on l_2, then $9(\alpha+\beta+\gamma)$ is equal to _____.

Open solution

Q181. JEE Main 2023 (Online) 6th April Evening Shift

If the lines $\frac{x-1}{2}=\frac{2-y}{-3}=\frac{z-3}{\alpha}$ and $\frac{x-4}{5}=\frac{y-1}{2}=\frac{z}{\beta}$ intersect, then the magnitude of the minimum value of $8\alpha\beta$ is _____.

Open solution

Q182. JEE Main 2023 (Online) 30th January Evening Shift

Let a line L pass through the point $P(2,3,1)$ and be parallel to the line $x+3y-2z-2=0=x-y+2z$. If the distance of L from the point $(5,3,8)$ is α, then $3\alpha^2$ is equal to :

Open solution

Q183. JEE Main 2023 (Online) 29th January Morning Shift

Let the co-ordinates of one vertex of ΔABC be $A(0,2,\alpha)$ and the other two vertices lie on the line $\frac{x+\alpha}{5}=\frac{y-1}{2}=\frac{z+4}{3}$. For $\alpha\in\mathbb{Z}$, if the area of ΔABC is 21 sq. units and the line segment BC has length $2\sqrt{21}$ units, then α^2 is equal to _____.

Open solution

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Q184. JEE Main 2023 (Online) 25th January Evening Shift

If the shortest distance between the line joining the points (1, 2, 3) and (2, 3, 4), and the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-2}{0}$ is α, then $28\alpha^2$ is equal to _____.

Open solution

Q185. JEE Main 2023 (Online) 24th January Evening Shift

If the shortest distance between the lines $\frac{x+\sqrt{6}}{2} = \frac{y-\sqrt{6}}{3} = \frac{z-\sqrt{6}}{4}$ and $\frac{x-\lambda}{3} = \frac{y-2\sqrt{6}}{4} = \frac{z+2\sqrt{6}}{5}$ is 6, then the square of sum of all possible values of λ is :

Open solution

Q186. JEE Main 2023 (Online) 24th January Morning Shift

The shortest distance between the lines $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-6}{2}$ and $\frac{x-6}{3} = \frac{1-y}{2} = \frac{z+8}{0}$ is equal to _____

Open solution

Q187. JEE Main 2022 (Online) 30th July Evening Shift

Let L_1 be a line in the yz-plane with y and z intercepts $\frac{1}{4}$ and $\frac{1}{C}$ $(C > 0)$ respectively. Let L_2 be the line in the xz-plane with x and z intercepts $\frac{1}{3}$ and $-\frac{1}{C}$ respectively. If the shortest distance between L_1 and L_2 is $\frac{1}{5}$, then C^2 is equal to _____.

Open solution

Q188. JEE Main 2022 (Online) 28th July Morning Shift

Let $\mathrm{P}(-2, -1, 1)$ and $\mathrm{Q}\left(\frac{56}{17}, \frac{43}{17}, \frac{111}{17}\right)$ be the vertices of the rhombus PRQS. If the direction ratios of the diagonal RS are $\alpha, -1, \beta$, where both α and β are integers of minimum absolute values, then $\alpha^2 + \beta^2$ is equal to _____.

Open solution

Q189. JEE Main 2022 (Online) 26th July Morning Shift

Let Q and R be two points on the line $\frac{x+1}{2} = \frac{y+2}{3} = \frac{z-1}{2}$ at a distance $\sqrt{26}$ from the point $P(4, 2, 7)$. Then the square of the area of the triangle PQR is _____.

Open solution

Q190. JEE Main 2022 (Online) 30th June Morning Shift

Consider a triangle ABC whose vertices are A(0, α, α), B(α, 0, α) and C(α, α, 0), $\alpha > 0$. Let D be a point moving on the line x + z − 3 = 0 = y and G be the centroid of ΔABC. If the minimum length of GD is $\sqrt{\frac{57}{2}}$, then α is equal to _____.

Open solution

Q191. JEE Main 2022 (Online) 28th June Evening Shift

Let the image of the point P(1, 2, 3) in the line $L : \frac{x-6}{3} = \frac{y-1}{2} = \frac{z-2}{3}$ be Q. Let R (α, β, γ) be a point that divides internally the line segment PQ in the ratio 1 : 3. Then the value of 22 $(\alpha + \beta + \gamma)$ is equal to _____.

Open solution

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Q192. JEE Main 2022 (Online) 25th June Evening Shift

Let l_1 be the line in xy-plane with x and y intercepts $\frac{1}{8}$ and $\frac{1}{4\sqrt{2}}$ respectively, and l_2 be the line in zx-plane with x and z intercepts $-\frac{1}{8}$ and $-\frac{1}{6\sqrt{3}}$ respectively. If d is the shortest distance between the line l_1 and l_2, then d^{-2} is equal to _____.

Open solution

Q193. JEE Main 2022 (Online) 24th June Morning Shift

Let a line having direction ratios, 1, −4, 2 intersect the lines $\frac{x-7}{3} = \frac{y-1}{-1} = \frac{z+2}{1}$ and $\frac{x}{2} = \frac{y-7}{3} = \frac{z}{1}$ at the points A and B. Then $(AB)^2$ is equal to _____.

Open solution

Q194. JEE Main 2022 (Online) 24th June Morning Shift

If the shortest distance between the lines

$\vec{r} = \left(-\hat{i} + 3\hat{k}\right) + \lambda\left(\hat{i} - a\hat{j}\right)$

and $\vec{r} = \left(-\hat{j} + 2\hat{k}\right) + \mu\left(\hat{i} - \hat{j} + \hat{k}\right)$ is $\sqrt{\frac{2}{3}}$, then the integral value of a is equal to _____.

Open solution

Q195. JEE Main 2021 (Online) 20th July Morning Shift

If the shortest distance between the lines $\vec{r_1} = \alpha\hat{i} + 2\hat{j} + 2\hat{k} + \lambda(\hat{i} - 2\hat{j} + 2\hat{k})$, $\lambda \in$ R, α > 0 and $\vec{r_2} = -4\hat{i} - \hat{k} + \mu(3\hat{i} - 2\hat{j} - 2\hat{k})$, $\mu \in$ R is 9, then α is equal to _____.

Open solution

Q196. JEE Main 2021 (Online) 25th February Evening Shift

A line 'l' passing through origin is perpendicular to the lines

$l_1 : \vec{r} = (3 + t)\hat{i} + (-1 + 2t)\hat{j} + (4 + 2t)\hat{k}$

$l_2 : \vec{r} = (3 + 2s)\hat{i} + (3 + 2s)\hat{j} + (2 + s)\hat{k}$

If the co-ordinates of the point in the first octant on 'l_2' at a distance of $\sqrt{17}$ from the point of intersection of 'l' and 'l_1' are (a, b, c) then 18(a + b + c) is equal to _____.

Open solution

Q197. JEE Main 2021 (Online) 24th February Evening Shift

Let λ be an integer. If the shortest distance between the lines

$x - \lambda = 2y - 1 = -2z$ and $x = y + 2\lambda = z - \lambda$ is $\frac{\sqrt{7}}{2\sqrt{2}}$, then the value of $|\lambda|$ is _____.

Open solution

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Q198. JEE Main 2020 (Online) 9th January Morning Slot

The projection of the line segment joining the points (1, −1, 3) and (2, −4, 11) on the line joining the points (−1, 2, 3) and (3, −2, 10) is _____.

Open solution

Q199. JEE Main 2020 (Online) 7th January Evening Slot

If the foot of the perpendicular drawn from the point (1, 0, 3) on a line passing through $(\alpha, 7, 1)$ is $\left(\frac{5}{3}, \frac{7}{3}, \frac{17}{3}\right)$, then α is equal to_____.

Open solution

Out of syllabus 219 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q200. JEE Main 2023 (Online) 15th April Morning Shift · MCQ (Single Correct Answer)

Let the foot of perpendicular of the point $P(3, -2, -9)$ on the plane passing through the points $(-1, -2, -3), (9, 3, 4), (9, -2, 1)$ be $Q(\alpha, \beta, \gamma)$. Then the distance of Q from the origin is :

(A) $\sqrt{38}$ (B) $\sqrt{29}$ (C) $\sqrt{42}$ (D) $\sqrt{35}$

Open solution

Q201. JEE Main 2023 (Online) 13th April Evening Shift · MCQ (Single Correct Answer)

The plane, passing through the points $(0, -1, 2)$ and $(-1, 2, 1)$ and parallel to the line passing through $(5, 1, -7)$ and $(1, -1, -1)$, also passes through the point :

(A) $(0, 5, -2)$ (B) $(2, 0, 1)$ (C) $(1, -2, 1)$ (D) $(-2, 5, 0)$

Open solution

Q202. JEE Main 2023 (Online) 13th April Evening Shift · MCQ (Single Correct Answer)

Let N be the foot of perpendicular from the point $\mathrm{P}(1, -2, 3)$ on the line passing through the points $(4, 5, 8)$ and $(1, -7, 5)$. Then the distance of N from the plane $2x - 2y + z + 5 = 0$ is :

(A) 7 (B) 6 (C) 9 (D) 8

Open solution

Q203. JEE Main 2023 (Online) 13th April Morning Shift · MCQ (Single Correct Answer)

Let the equation of plane passing through the line of intersection of the planes $x + 2y + az = 2$ and $x - y + z = 3$ be $5x - 11y + bz = 6a - 1$. For $c \in \mathbb{Z}$, if the distance of this plane from the point $(a, -c, c)$ is $\frac{2}{\sqrt{a}}$, then $\frac{a+b}{c}$ is equal to :

(A) -2 (B) 4 (C) 2 (D) -4

Open solution

Q204. JEE Main 2023 (Online) 13th April Morning Shift · MCQ (Single Correct Answer)

The distance of the point $(-1, 2, 3)$ from the plane $\vec{r} \cdot (\hat{i} - 2\hat{j} + 3\hat{k}) = 10$ parallel to the line of the shortest distance between the lines $\vec{r} = (\hat{i} - \hat{j}) + \lambda(2\hat{i} + \hat{k})$ and $\vec{r} = (2\hat{i} - \hat{j}) + \mu(\hat{i} - \hat{j} + \hat{k})$ is :

(A) $3\sqrt{6}$ (B) $3\sqrt{5}$ (C) $2\sqrt{6}$ (D) $2\sqrt{5}$

Open solution

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Q205. JEE Main 2023 (Online) 12th April Morning Shift · MCQ (Single Correct Answer)

Let the plane P: $4x - y + z = 10$ be rotated by an angle $\frac{\pi}{2}$ about its line of intersection with the plane $x + y - z = 4$. If α is the distance of the point $(2, 3, -4)$ from the new position of the plane P, then 35α is equal to :

Open solution

(A) 126 (B) 105 (C) 85 (D) 90

Q206. JEE Main 2023 (Online) 11th April Evening Shift · MCQ (Single Correct Answer)

Let the line passing through the points $\mathrm{P}(2, -1, 2)$ and $\mathrm{Q}(5, 3, 4)$ meet the plane $x - y + z = 4$ at the point R. Then the distance of the point R from the plane $x + 2y + 3z + 2 = 0$ measured parallel to the line $\frac{x-7}{2} = \frac{y+3}{2} = \frac{z-2}{1}$ is equal to :

Open solution

(A) $\sqrt{31}$ (B) $\sqrt{189}$ (C) $\sqrt{61}$ (D) 3

Q207. JEE Main 2023 (Online) 11th April Evening Shift · MCQ (Single Correct Answer)

Let P be the plane passing through the points $(5, 3, 0), (13, 3, -2)$ and $(1, 6, 2)$. For $\alpha \in \mathbb{N}$, if the distances of the points $\mathrm{A}(3, 4, \alpha)$ and $\mathrm{B}(2, \alpha, a)$ from the plane P are 2 and 3 respectively, then the positive value of a is :

Open solution

(A) 6 (B) 4 (C) 5 (D) 3

Q208. JEE Main 2023 (Online) 11th April Morning Shift · MCQ (Single Correct Answer)

Let (α, β, γ) be the image of the point $\mathrm{P}(2, 3, 5)$ in the plane $2x + y - 3z = 6$. Then $\alpha + \beta + \gamma$ is equal to :

Open solution

(A) 10 (B) 9 (C) 5 (D) 12

Q209. JEE Main 2023 (Online) 11th April Morning Shift · MCQ (Single Correct Answer)

If equation of the plane that contains the point $(-2, 3, 5)$ and is perpendicular to each of the planes $2x + 4y + 5z = 8$ and $3x - 2y + 3z = 5$ is $\alpha x + \beta y + \gamma z + 97 = 0$ then $\alpha + \beta + \gamma =$

Open solution

(A) 15 (B) 16 (C) 17 (D) 18

Q210. JEE Main 2023 (Online) 10th April Evening Shift · MCQ (Single Correct Answer)

Let the image of the point $\mathrm{P}(1, 2, 6)$ in the plane passing through the points $\mathrm{A}(1, 2, 0), \mathrm{B}(1, 4, 1)$ and $\mathrm{C}(0, 5, 1)$ be $\mathrm{Q}(\alpha, \beta, \gamma)$. Then $(\alpha^2 + \beta^2 + \gamma^2)$ is equal to :

Open solution

(A) 76 (B) 62 (C) 70 (D) 65

Q211. JEE Main 2023 (Online) 10th April Evening Shift · MCQ (Single Correct Answer)

Let the line $\frac{x}{1} = \frac{6-y}{2} = \frac{z+8}{5}$ intersect the lines $\frac{x-5}{4} = \frac{y-7}{3} = \frac{z+2}{1}$ and $\frac{x+3}{6} = \frac{3-y}{3} = \frac{z-6}{1}$ at the points A and B respectively. Then the distance of the mid-point of the line segment AB from the plane $2x - 2y + z = 14$ is :

Open solution

(A) 3 (B) $\frac{10}{3}$ (C) 4 (D) $\frac{11}{3}$

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Q212. JEE Main 2023 (Online) 10th April Morning Shift · MCQ (Single Correct Answer)

Let two vertices of a triangle ABC be (2, 4, 6) and (0, −2, −5), and its centroid be (2, 1, −1). If the image of the third vertex in the plane $x + 2y + 4z = 11$ is (α, β, γ), then $\alpha\beta + \beta\gamma + \gamma\alpha$ is equal to :

(A) 72 (B) 74 (C) 76 (D) 70

Open solution

Q213. JEE Main 2023 (Online) 10th April Morning Shift · MCQ (Single Correct Answer)

Let P be the point of intersection of the line $\frac{x+3}{3} = \frac{y+2}{1} = \frac{1-z}{2}$ and the plane $x + y + z = 2$. If the distance of the point P from the plane $3x - 4y + 12z = 32$ is q, then q and 2q are the roots of the equation :

(A) $x^2 + 18x - 72 = 0$ (B) $x^2 - 18x - 72 = 0$ (C) $x^2 + 18x + 72 = 0$ (D) $x^2 - 18x + 72 = 0$

Open solution

Q214. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

For $\mathrm{a}, \mathrm{b} \in \mathbb{Z}$ and $|\mathrm{a} - \mathrm{b}| \leq 10$, let the angle between the plane $\mathrm{P} : \mathrm{ax} + y - \mathrm{z} = \mathrm{b}$ and the line $l : x - 1 = \mathrm{a} - y = z + 1$ be $\cos^{-1}\left(\frac{1}{3}\right)$. If the distance of the point $(6, -6, 4)$ from the plane P is $3\sqrt{6}$, then $a^4 + b^2$ is equal to :

(A) 48 (B) 85 (C) 32 (D) 25

Open solution

Q215. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

Let P be the plane passing through the line

$\frac{x-1}{1} = \frac{y-2}{-3} = \frac{z+5}{7}$ and the point $(2, 4, -3)$.

If the image of the point $(-1, 3, 4)$ in the plane P

is (α, β, γ) then $\alpha + \beta + \gamma$ is equal to :

(A) 10 (B) 12 (C) 9 (D) 11

Open solution

Q216. JEE Main 2023 (Online) 8th April Morning Shift · MCQ (Single Correct Answer)

If the equation of the plane containing the line

$x + 2y + 3z - 4 = 0 = 2x + y - z + 5$ and perpendicular to the plane

$\vec{r} = (\hat{i} - \hat{j}) + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$

is $ax + by + cz = 4$, then $(a - b + c)$ is equal to :

(A) 18 (B) 22 (C) 20 (D) 24

Open solution

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Q217. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

A plane P contains the line of intersection of the plane $\vec{r} \cdot (\hat{i} + \hat{j} + \hat{k}) = 6$ and $\vec{r} \cdot (2\hat{i} + 3\hat{j} + 4\hat{k}) = -5$. If P passes through the point $(0, 2, -2)$, then the square of distance of the point $(12, 12, 18)$ from the plane P is :

Open solution

(A) 310 (B) 620 (C) 1240 (D) 155

Q218. JEE Main 2023 (Online) 6th April Evening Shift · MCQ (Single Correct Answer)

Let the line L pass through the point $(0, 1, 2)$, intersect the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and be parallel to the plane $2x + y - 3z = 4$. Then the distance of the point $\mathrm{P}(1, -9, 2)$ from the line L is :

Open solution

(A) 9 (B) $\sqrt{74}$ (C) $\sqrt{69}$ (D) $\sqrt{54}$

Q219. JEE Main 2023 (Online) 6th April Morning Shift · MCQ (Single Correct Answer)

If the equation of the plane passing through the line of intersection of the planes $2x - y + z = 3, 4x - 3y + 5z + 9 = 0$ and parallel to the line $\frac{x+1}{-2} = \frac{y+3}{4} = \frac{z-2}{5}$ is $ax + by + cz + 6 = 0$, then $a + b + c$ is equal to :

Open solution

(A) 13 (B) 15 (C) 14 (D) 12

Q220. JEE Main 2023 (Online) 1st February Evening Shift · MCQ (Single Correct Answer)

Let the plane P pass through the intersection of the planes $2x + 3y - z = 2$ and $x + 2y + 3z = 6$, and be perpendicular to the plane $2x + y - z + 1 = 0$. If d is the distance of P from the point $(-7, 1, 1)$, then d^2 is equal to :

Open solution

(A) $\frac{250}{83}$ (B) $\frac{250}{82}$ (C) $\frac{15}{53}$ (D) $\frac{25}{83}$

Q221. JEE Main 2023 (Online) 1st February Morning Shift · MCQ (Single Correct Answer)

Let the image of the point $P(2, -1, 3)$ in the plane $x + 2y - z = 0$ be Q.

Then the distance of the plane $3x + 2y + z + 29 = 0$ from the point Q is :

Open solution

(A) $2\sqrt{14}$ (B) $\frac{22\sqrt{2}}{7}$ (C) $\frac{24\sqrt{2}}{7}$ (D) $3\sqrt{14}$

Q222. JEE Main 2023 (Online) 31st January Evening Shift · MCQ (Single Correct Answer)

Let the plane $\mathrm{P} : 8x + \alpha_1 y + \alpha_2 z + 12 = 0$ be parallel to

the line $\mathrm{L} : \frac{x+2}{2} = \frac{y-3}{3} = \frac{z+4}{5}$. If the intercept of P

on the y-axis is 1 , then the distance between P and L is :

Open solution

(A) $\frac{6}{\sqrt{14}}$ (B) $\sqrt{14}$ (C) $\sqrt{\frac{2}{7}}$ (D) $\sqrt{\frac{7}{2}}$

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Q223. JEE Main 2023 (Online) 31st January Evening Shift · MCQ (Single Correct Answer)

The foot of perpendicular from the origin O to a plane P which meets the co-ordinate axes at the points A, B, C is $(2, a, 4), a \in N$. If the volume of the tetrahedron OABC is 144 unit^3, then which of the following points is **NOT** on P ?

(A) $(3, 0, 4)$ (B) $(0, 6, 3)$ (C) $(0, 4, 4)$ (D) $(2, 2, 4)$

Open solution

Q224. JEE Main 2023 (Online) 31st January Evening Shift · MCQ (Single Correct Answer)

Let P be the plane, passing through the point $(1, -1, -5)$ and perpendicular to the line joining the points $(4, 1, -3)$ and $(2, 4, 3)$. Then the distance of P from the point $(3, -2, 2)$ is :

(A) 5 (B) 4 (C) 6 (D) 7

Open solution

Q225. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

A vector $\vec{v}$ in the first octant is inclined to the x-axis at $60°$, to the y-axis at 45 and to the z-axis at an acute angle. If a plane passing through the points $(\sqrt{2}, -1, 1)$ and (a, b, c), is normal to $\vec{v}$, then :

(A) $a + b + \sqrt{2}c = 1$ (B) $\sqrt{2}a + b + c = 1$ (C) $\sqrt{2}a - b + c = 1$ (D) $a + \sqrt{2}b + c = 1$

Open solution

Q226. JEE Main 2023 (Online) 30th January Evening Shift · MCQ (Single Correct Answer)

If a plane passes through the points $(-1, k, 0), (2, k, -1), (1, 1, 2)$ and is parallel to the line $\frac{x-1}{1} = \frac{2y+1}{2} = \frac{z+1}{-1}$, then the value of $\frac{k^2+1}{(k-1)(k-2)}$ is :

(A) $\frac{17}{5}$ (B) $\frac{6}{13}$ (C) $\frac{13}{6}$ (D) $\frac{5}{17}$

Open solution

Q227. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

The line l_1 passes through the point (2, 6, 2) and is perpendicular to the plane $2x + y - 2z = 10$. Then the shortest distance between the line l_1 and the line $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ is :

(A) 9 (B) 7 (C) $\frac{19}{3}$ (D) $\frac{13}{3}$

Open solution

Q228. JEE Main 2023 (Online) 29th January Evening Shift · MCQ (Single Correct Answer)

The plane $2x - y + z = 4$ intersects the line segment joining the points A $(a, -2, 4)$ and B $(2, b, -3)$ at the point C in the ratio 2 : 1 and the distance of the point C from the origin is $\sqrt{5}$. If $ab < 0$ and P is the point $(a - b, b, 2b - a)$ then CP^2 is equal to :

(A) $\frac{17}{3}$ (B) $\frac{97}{3}$ (C) $\frac{16}{3}$ (D) $\frac{73}{3}$

Open solution

Q229. JEE Main 2023 (Online) 29th January Evening Shift · MCQ (Single Correct Answer)

If the lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{1}$ and $\frac{x-a}{2} = \frac{y+2}{3} = \frac{z-3}{1}$ intersect at the point P, then the distance of the point P from the plane $z = a$ is :

(A) 28 (B) 22 (C) 10 (D) 16

Open solution

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Q230. JEE Main 2023 (Online) 24th January Evening Shift · MCQ (Single Correct Answer)

If the foot of the perpendicular drawn from (1, 9, 7) to the line passing through the point (3, 2, 1) and parallel to the planes $x + 2y + z = 0$ and $3y - z = 3$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to :

(A) 3 (B) 1 (C) -1 (D) 5

Open solution

Q231. JEE Main 2023 (Online) 24th January Evening Shift · MCQ (Single Correct Answer)

Let the plane containing the line of intersection of the planes

$P_1 : x + (\lambda + 4)y + z = 1$ and

$P_2 : 2x + y + z = 2$

pass through the points (0, 1, 0) and (1, 0, 1). Then the distance of the point $(2\lambda, \lambda, -\lambda)$ from the plane P_2 is :

(A) $2\sqrt{6}$ (B) $3\sqrt{6}$ (C) $4\sqrt{6}$ (D) $5\sqrt{6}$

Open solution

Q232. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

The distance of the point (7, -3, -4) from the plane passing through the points (2, -3, 1), (-1, 1, -2) and (3, -4, 2) is :

(A) $4\sqrt{2}$ (B) 4 (C) 5 (D) $5\sqrt{2}$

Open solution

Q233. JEE Main 2023 (Online) 24th January Morning Shift · MCQ (Single Correct Answer)

The distance of the point $(-1, 9, -16)$ from the plane

$2x + 3y - z = 5$ measured parallel to the line

$\frac{x+4}{3} = \frac{2-y}{4} = \frac{z-3}{12}$ is :

(A) $13\sqrt{2}$ (B) 26 (C) $20\sqrt{2}$ (D) 31

Open solution

Q234. JEE Main 2022 (Online) 30th July Evening Shift · MCQ (Single Correct Answer)

Let $ax + by + cz + 3 = 0$ be the plane which bisects the acute angle between the planes $2x - y - 2z + 3 = 0$ and $3x - 2y + 6z + 8 = 0$. Then $a + b + c$ is equal to

(A) -31 (B) 28 (C) $\frac{14}{15}$ (D) -28

Open solution

Q235. JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

Let Q be the foot of perpendicular drawn from the point $P(1, 2, 3)$ to the plane $x + 2y + z = 14$. If R is a point on the plane such that $\angle PRQ = 60^\circ$, then the area of $\triangle PQR$ is equal to :

(A) $\frac{\sqrt{3}}{2}$ (B) $\sqrt{3}$ (C) $2\sqrt{3}$ (D) 3

Open solution

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Q236. JEE Main 2022 (Online) 29th July Evening Shift · MCQ (Single Correct Answer)

If $(2,3,9),(5,2,1),(1,\lambda,8)$ and $(\lambda,2,3)$ are coplanar, then the product of all possible values of λ is:

(A) $\frac{21}{2}$ (B) $\frac{59}{8}$ (C) $\frac{57}{8}$ (D) $\frac{95}{8}$

Open solution

Q237. JEE Main 2022 (Online) 29th July Morning Shift · MCQ (Single Correct Answer)

If the foot of the perpendicular from the point $\mathrm{A}(-1,4,3)$ on the plane $\mathrm{P}: 2x+\mathrm{m}y+\mathrm{n}z=4$, is $\left(-2,\frac{7}{2},\frac{3}{2}\right)$, then the distance of the point A from the plane P, measured parallel to a line with direction ratios $3,-1,-4$, is equal to :

(A) 1 (B) $\sqrt{26}$ (C) $2\sqrt{2}$ (D) $\sqrt{14}$

Open solution

Q238. JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

Let the lines

$\frac{x-1}{\lambda}=\frac{y-2}{1}=\frac{z-3}{2}$ and

$\frac{x+26}{-2}=\frac{y+18}{3}=\frac{z+28}{\lambda}$ be coplanar

and P be the plane containing these two lines.

Then which of the following points does **NOT** lie on P?

(A) $(0,-2,-2)$ (B) $(-5,0,-1)$ (C) $(3,-1,0)$ (D) $(0,4,5)$

Open solution

Q239. JEE Main 2022 (Online) 28th July Evening Shift · MCQ (Single Correct Answer)

A plane P is parallel to two lines whose direction ratios are $-2,1,-3$ and $-1,2,-2$ and it contains the point $(2,2,-2)$. Let P intersect the co-ordinate axes at the points $\mathrm{A},\mathrm{B},\mathrm{C}$ making the intercepts α,β,γ. If V is the volume of the tetrahedron OABC, where O is the origin, and $\mathrm{p}=\alpha+\beta+\gamma$, then the ordered pair (V,p) is equal to :

(A) $(48,-13)$ (B) $(24,-13)$ (C) $(48,11)$ (D) $(24,-5)$

Open solution

Q240. JEE Main 2022 (Online) 28th July Morning Shift · MCQ (Single Correct Answer)

The foot of the perpendicular from a point on the circle $x^2+y^2=1, z=0$ to the plane $2x+3y+z=6$ lies on which one of the following curves?

(A) $(6x+5y-12)^2+4(3x+7y-8)^2=1, z=6-2x-3y$

(B) $(5x+6y-12)^2+4(3x+5y-9)^2=1, z=6-2x-3y$

(C) $(6x+5y-14)^2+9(3x+5y-7)^2=1, z=6-2x-3y$

(D) $(5x+6y-14)^2+9(3x+7y-8)^2=1, z=6-2x-3y$

Open solution

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Q241. JEE Main 2022 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

If the line of intersection of the planes $ax + by = 3$ and $ax + by + cz = 0$, a > 0 makes an angle 30° with the plane $y - z + 2 = 0$, then the direction cosines of the line are :

(A) $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$

(B) $\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}, 0$

(C) $\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, 0$

(D) $\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0$

Open solution

Q242. JEE Main 2022 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

If the plane P passes through the intersection of two mutually perpendicular planes $2x + ky - 5z = 1$ and $3kx - ky + z = 5, k < 3$ and intercepts a unit length on positive x-axis, then the intercept made by the plane P on the y-axis is :

(A) $\frac{1}{11}$

(B) $\frac{5}{11}$

(C) 6

(D) 7

Open solution

Q243. JEE Main 2022 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

A plane E is perpendicular to the two planes $2x - 2y + z = 0$ and $x - y + 2z = 4$, and passes through the point $P(1, -1, 1)$. If the distance of the plane E from the point $Q(a, a, 2)$ is $3\sqrt{2}$, then $(PQ)^2$ is equal to :

(A) 9

(B) 12

(C) 21

(D) 33

Open solution

Q244. JEE Main 2022 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

Let P be the plane containing the straight line $\frac{x-3}{9} = \frac{y+4}{-1} = \frac{z-7}{-5}$ and perpendicular to the plane containing the straight lines $\frac{x}{2} = \frac{y}{3} = \frac{z}{5}$ and $\frac{x}{3} = \frac{y}{7} = \frac{z}{8}$. If d is the distance of P from the point $(2, -5, 11)$, then d^2 is equal to :

(A) $\frac{147}{2}$

(B) 96

(C) $\frac{32}{3}$

(D) 54

Open solution

Q245. JEE Main 2022 (Online) 30th June Morning Shift · MCQ (Single Correct Answer)

The distance of the point (3, 2, -1) from the plane $3x - y + 4z + 1 = 0$ along the line $\frac{2-x}{2} = \frac{y-3}{2} = \frac{z+1}{1}$ is equal to :

(A) 9

(B) 6

(C) 3

(D) 2

Open solution

Q246. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Let $\frac{x-2}{3} = \frac{y+1}{-2} = \frac{z+3}{-1}$ lie on the plane $px - qy + z = 5$, for some p, q $\in$ R. The shortest distance of the plane from the origin is :

(A) $\sqrt{\frac{3}{109}}$

(B) $\sqrt{\frac{5}{142}}$

(C) $\frac{5}{\sqrt{71}}$

(D) $\frac{1}{\sqrt{142}}$

Open solution

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Q247. JEE Main 2022 (Online) 29th June Evening Shift · MCQ (Single Correct Answer)

Let Q be the mirror image of the point P(1, 2, 1) with respect to the plane x + 2y + 2z = 16. Let T be a plane passing through the point Q and contains the line $\vec{r} = -\hat{k} + \lambda\left(\hat{i} + \hat{j} + 2\hat{k}\right)$, $\lambda \in R$. Then, which of the following points lies on T?

Open solution

(A) (2, 1, 0) (B) (1, 2, 1) (C) (1, 2, 2) (D) (1, 3, 2)

Q248. JEE Main 2022 (Online) 29th June Morning Shift · MCQ (Single Correct Answer)

If the mirror image of the point (2, 4, 7) in the plane 3x − y + 4z = 2 is (a, b, c), then 2a + b + 2c is equal to :

Open solution

(A) 54 (B) 50 (C) −6 (D) −42

Q249. JEE Main 2022 (Online) 28th June Evening Shift · MCQ (Single Correct Answer)

Let the plane ax + by + cz = d pass through (2, 3, −5) and is perpendicular to the planes 2x + y − 5z = 10 and 3x + 5y − 7z = 12. If a, b, c, d are integers d > 0 and gcd (|a|, |b|, |c|, d) = 1, then the value of a + 7b + c + 20d is equal to :

Open solution

(A) 18 (B) 20 (C) 24 (D) 22

Q250. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

If two distinct point Q, R lie on the line of intersection of the planes $-x + 2y - z = 0$ and $3x - 5y + 2z = 0$ and $PQ = PR = \sqrt{18}$ where the point P is (1, −2, 3), then the area of the triangle PQR is equal to :

Open solution

(A) $\frac{2}{3}\sqrt{38}$ (B) $\frac{4}{3}\sqrt{38}$ (C) $\frac{8}{3}\sqrt{38}$ (D) $\sqrt{\frac{152}{3}}$

Q251. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

The acute angle between the planes P_1 and P_2, when P_1 and P_2 are the planes passing through the intersection of the planes $5x + 8y + 13z - 29 = 0$ and $8x - 7y + z - 20 = 0$ and the points (2, 1, 3) and (0, 1, 2), respectively, is :

Open solution

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

Q252. JEE Main 2022 (Online) 28th June Morning Shift · MCQ (Single Correct Answer)

Let the plane $P : \vec{r}.\vec{a} = d$ contain the line of intersection of two planes $\vec{r}.\left(\hat{i} + 3\hat{j} - \hat{k}\right) = 6$ and $\vec{r}.\left(-6\hat{i} + 5\hat{j} - \hat{k}\right) = 7$. If the plane P passes through the point $\left(2, 3, \frac{1}{2}\right)$, then the value of $\frac{|13\vec{a}|^2}{d^2}$ is equal to :

Open solution

(A) 90 (B) 93 (C) 95 (D) 97

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Q253. JEE Main 2022 (Online) 27th June Evening Shift · MCQ (Single Correct Answer)

Let the foot of the perpendicular from the point (1, 2, 4) on the line $\frac{x+2}{4} = \frac{y-1}{2} = \frac{z+1}{3}$ be P. Then the distance of P from the plane $3x + 4y + 12z + 23 = 0$ is :

(A) 5 (B) $\frac{50}{13}$ (C) 4 (D) $\frac{63}{13}$

Open solution

Q254. JEE Main 2022 (Online) 26th June Evening Shift · MCQ (Single Correct Answer)

If the plane $2x + y - 5z = 0$ is rotated about its line of intersection with the plane $3x - y + 4z - 7 = 0$ by an angle of $\frac{\pi}{2}$, then the plane after the rotation passes through the point :

(A) (2, −2, 0) (B) (−2, 2, 0) (C) (1, 0, 2) (D) (−1, 0, −2)

Open solution

Q255. JEE Main 2022 (Online) 26th June Evening Shift · MCQ (Single Correct Answer)

If the lines $\vec{r} = \left(\hat{i} - \hat{j} + \hat{k}\right) + \lambda\left(3\hat{j} - \hat{k}\right)$ and $\vec{r} = \left(\alpha\hat{i} - \hat{j}\right) + \mu\left(2\hat{i} - 3\hat{k}\right)$ are co-planar, then the distance of the plane containing these two lines from the point $(\alpha, 0, 0)$ is :

(A) $\frac{2}{9}$ (B) $\frac{2}{11}$ (C) $\frac{4}{11}$ (D) 2

Open solution

Q256. JEE Main 2022 (Online) 26th June Morning Shift · MCQ (Single Correct Answer)

Let the plane 2x + 3y + z + 20 = 0 be rotated through a right angle about its line of intersection with the plane x − 3y + 5z = 8. If the mirror image of the point $\left(2, -\frac{1}{2}, 2\right)$ in the rotated plane is B(a, b, c), then :

(A) $\frac{a}{8} = \frac{b}{5} = \frac{c}{-4}$ (B) $\frac{a}{4} = \frac{b}{5} = \frac{c}{-2}$

(C) $\frac{a}{8} = \frac{b}{-5} = \frac{c}{4}$ (D) $\frac{a}{4} = \frac{b}{5} = \frac{c}{2}$

Open solution

Q257. JEE Main 2022 (Online) 25th June Evening Shift · MCQ (Single Correct Answer)

Let p be the plane passing through the intersection of the planes $\vec{r}.\left(\hat{i} + 3\hat{j} - \hat{k}\right) = 5$ and $\vec{r}.\left(2\hat{i} - \hat{j} + \hat{k}\right) = 3$, and the point (2, 1, −2). Let the position vectors of the points X and Y be $\hat{i} - 2\hat{j} + 4\hat{k}$ and $5\hat{i} - \hat{j} + 2\hat{k}$ respectively. Then the points :

(A) X and X + Y are on the same side of P

(B) Y and Y − X are on the opposite sides of P

(C) X and Y are on the opposite sides of P

(D) X + Y and X − Y are on the same side of P

Open solution

Q258. JEE Main 2022 (Online) 25th June Morning Shift · MCQ (Single Correct Answer)

Let Q be the mirror image of the point P(1, 0, 1) with respect to the plane S : x + y + z = 5. If a line L passing through (1, −1, −1), parallel to the line PQ meets the plane S at R, then QR^2 is equal to :

(A) 2 (B) 5 (C) 7 (D) 11

Open solution

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Q259. JEE Main 2022 (Online) 24th June Evening Shift · MCQ (Single Correct Answer)

Let the points on the plane P be equidistant from the points $(-4, 2, 1)$ and $(2, -2, 3)$. Then the acute angle between the plane P and the plane $2x + y + 3z = 1$ is :

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{5\pi}{12}$

Open solution

Q260. JEE Main 2021 (Online) 1st September Evening Shift · MCQ (Single Correct Answer)

Let the acute angle bisector of the two planes $x - 2y - 2z + 1 = 0$ and $2x - 3y - 6z + 1 = 0$ be the plane P. Then which of the following points lies on P?

(A) $\left(3, 1, -\frac{1}{2}\right)$ (B) $\left(-2, 0, -\frac{1}{2}\right)$

(C) $(0, 2, -4)$ (D) $(4, 0, -2)$

Open solution

Q261. JEE Main 2021 (Online) 31st August Evening Shift · MCQ (Single Correct Answer)

The distance of the point $(-1, 2, -2)$ from the line of intersection of the planes $2x + 3y + 2z = 0$ and $x - 2y + z = 0$ is :

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{5}{2}$ (C) $\frac{\sqrt{42}}{2}$ (D) $\frac{\sqrt{34}}{2}$

Open solution

Q262. JEE Main 2021 (Online) 31st August Morning Shift · MCQ (Single Correct Answer)

Let the equation of the plane, that passes through the point $(1, 4, -3)$ and contains the line of intersection of the
planes $3x - 2y + 4z - 7 = 0$
and $x + 5y - 2z + 9 = 0$, be
$\alpha x + \beta y + \gamma z + 3 = 0$, then $\alpha + \beta + \gamma$ is equal to :

(A) -23 (B) -15 (C) 23 (D) 15

Open solution

Q263. JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

The equation of the plane passing through the line of intersection of the planes $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(2\hat{i}+3\hat{j}-\hat{k}\right)+4=0$ and parallel to the x-axis is :

(A) $\vec{r}.\left(\hat{j}-3\hat{k}\right)+6=0$ (B) $\vec{r}.\left(\hat{i}+3\hat{k}\right)+6=0$

(C) $\vec{r}.\left(\hat{i}-3\hat{k}\right)+6=0$ (D) $\vec{r}.\left(\hat{j}-3\hat{k}\right)-6=0$

Open solution

Q264. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

The distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to a line, whose direction ratios are 2, 3, -6 is :

(A) 3 (B) 5 (C) 2 (D) 1

Open solution

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Q265. JEE Main 2021 (Online) 27th August Morning Shift · MCQ (Single Correct Answer)

Equation of a plane at a distance $\sqrt{\frac{2}{21}}$ from the origin, which contains the line of intersection of the planes x − y − z − 1 = 0 and 2x + y − 3z + 4 = 0, is :

(A) $3x - y - 5z + 2 = 0$ (B) $3x - 4z + 3 = 0$

(C) $-x + 2y + 2z - 3 = 0$ (D) $4x - y - 5z + 2 = 0$

Open solution

Q266. JEE Main 2021 (Online) 26th August Evening Shift · MCQ (Single Correct Answer)

Let P be the plane passing through the point (1, 2, 3) and the line of intersection of the planes $\vec{r}.\left(\hat{i}+\hat{j}+4\hat{k}\right)=16$ and $\vec{r}.\left(-\hat{i}+\hat{j}+\hat{k}\right)=6$. Then which of the following points does NOT lie on P?

(A) (3, 3, 2) (B) (6, −6, 2) (C) (4, 2, 2) (D) (−8, 8, 6)

Open solution

Q267. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

A plane P contains the line $x + 2y + 3z + 1 = 0 = x - y - z - 6$, and is perpendicular to the plane $-2x + y + z + 8 = 0$. Then which of the following points lies on P?

(A) (−1, 1, 2) (B) (0, 1, 1) (C) (1, 0, 1) (D) (2, −1, 1)

Open solution

Q268. JEE Main 2021 (Online) 27th July Evening Shift · MCQ (Single Correct Answer)

For real numbers α and $\beta \neq 0$, if the point of intersection of the straight lines

$\frac{x-\alpha}{1}=\frac{y-1}{2}=\frac{z-1}{3}$ and $\frac{x-4}{\beta}=\frac{y-6}{3}=\frac{z-7}{3}$, lies on the plane x + 2y − z = 8, then $\alpha - \beta$ is equal to :

(A) 5 (B) 9 (C) 3 (D) 7

Open solution

Q269. JEE Main 2021 (Online) 27th July Morning Shift · MCQ (Single Correct Answer)

Let the plane passing through the point (−1, 0, −2) and perpendicular to each of the planes 2x + y − z = 2 and x − y − z = 3 be ax + by + cz + 8 = 0. Then the value of a + b + c is equal to :

(A) 3 (B) 8 (C) 5 (D) 4

Open solution

Q270. JEE Main 2021 (Online) 25th July Morning Shift · MCQ (Single Correct Answer)

Let the foot of perpendicular from a point P(1, 2, −1) to the straight line $L: \frac{x}{1}=\frac{y}{0}=\frac{z}{-1}$ be N. Let a line be drawn from P parallel to the plane x + y + 2z = 0 which meets L at point Q. If α is the acute angle between the lines PN and PQ, then cosα is equal to _____.

(A) $\frac{1}{\sqrt{5}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{2\sqrt{3}}$

Open solution

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Q271. JEE Main 2021 (Online) 22th July Evening Shift · MCQ (Single Correct Answer)

Let L be the line of intersection of planes $\vec{r}.(\hat{i}-\hat{j}+2\hat{k})=2$ and $\vec{r}.(2\hat{i}+\hat{j}-\hat{k})=2$. If $P(\alpha,\beta,\gamma)$ is the foot of perpendicular on L from the point (1, 2, 0), then the value of $35(\alpha+\beta+\gamma)$ is equal to :

(A) 101 (B) 119 (C) 143 (D) 134

Open solution

Q272. JEE Main 2021 (Online) 20th July Evening Shift · MCQ (Single Correct Answer)

Consider the line L given by the equation

$\frac{x-3}{2}=\frac{y-1}{1}=\frac{z-2}{1}$.

Let Q be the mirror image of the point (2, 3, -1) with respect to L. Let a plane P be such that it passes through Q, and the line L is perpendicular to P. Then which of the following points is on the plane P?

(A) $(-1, 1, 2)$ (B) (1, 1, 1) (C) (1, 1, 2) (D) (1, 2, 2)

Open solution

Q273. JEE Main 2021 (Online) 17th March Evening Shift · MCQ (Single Correct Answer)

If the equation of plane passing through the mirror image of a point (2, 3, 1) with respect to line $\frac{x+1}{2}=\frac{y-3}{1}=\frac{z+2}{-1}$ and containing the line $\frac{x-2}{3}=\frac{1-y}{2}=\frac{z+1}{1}$ is $\alpha x+\beta y+\gamma z=24$, then $\alpha+\beta+\gamma$ is equal to :

(A) 21 (B) 19 (C) 18 (D) 20

Open solution

Q274. JEE Main 2021 (Online) 17th March Morning Shift · MCQ (Single Correct Answer)

The equation of the plane which contains the y-axis and passes through the point (1, 2, 3) is :

(A) $x+3z=0$ (B) $3x-z=0$ (C) $x+3z=10$ (D) $3x+z=6$

Open solution

Q275. JEE Main 2021 (Online) 16th March Evening Shift · MCQ (Single Correct Answer)

If (x, y, z) be an arbitrary point lying on a plane P which passes through the points (42, 0, 0), (0, 42, 0) and (0, 0, 42), then the value of the expression

$3+\frac{x-11}{(y-19)^2(z-12)^2}+\frac{y-19}{(x-11)^2(z-12)^2}+\frac{z-12}{(x-11)^2(y-19)^2}-\frac{x+y+z}{14(x-11)(y-19)(z-12)}$ is equal to :

(A) 3 (B) 39 (C) -45 (D) 0

Open solution

Q276. JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

Let P be a plane $lx+my+nz=0$ containing

the line, $\frac{1-x}{1}=\frac{y+4}{2}=\frac{z+2}{3}$. If plane P divides the line segment AB joining

points A(-3, -6, 1) and B(2, 4, -3) in ratio k : 1 then the value of k is equal to :

(A) 2 (B) 3 (C) 1.5 (D) 4

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Q277. JEE Main 2021 (Online) 16th March Morning Shift · MCQ (Single Correct Answer)

If for a > 0, the feet of perpendiculars from the points A(a, $-2a$, 3) and B(0, 4, 5) on the plane lx + my + nz = 0 are points C(0, $-a$, -1) and D respectively, then the length of line segment CD is equal to :

(A) $\sqrt{41}$ (B) $\sqrt{55}$ (C) $\sqrt{31}$ (D) $\sqrt{66}$

Open solution

Q278. JEE Main 2021 (Online) 26th February Evening Shift · MCQ (Single Correct Answer)

If the mirror image of the point (1, 3, 5) with respect to the plane

$4x - 5y + 2z = 8$ is (α, β, γ), then $5(\alpha + \beta + \gamma)$ equals :

(A) 39 (B) 41 (C) 47 (D) 43

Open solution

Q279. JEE Main 2021 (Online) 26th February Evening Shift · MCQ (Single Correct Answer)

Let L be a line obtained from the intersection of two planes x + 2y + z = 6 and y + 2z = 4. If point P(α, β, γ) is the foot of perpendicular from (3, 2, 1) on L, then the value of $21(\alpha + \beta + \gamma)$ equals :

(A) 102 (B) 142 (C) 136 (D) 68

Open solution

Q280. JEE Main 2021 (Online) 26th February Morning Shift · MCQ (Single Correct Answer)

Consider the three planes

$P_1 : 3x + 15y + 21z = 9$,

$P_2 : x - 3y - z = 5$, and

$P_3 : 2x + 10y + 14z = 5$

Then, which one of the following is true?

(A) P_1 and P_2 are parallel. (B) P_1, P_2 and P_3 all are parallel.

(C) P_1 and P_3 are parallel. (D) P_2 and P_3 are parallel.

Open solution

Q281. JEE Main 2021 (Online) 26th February Morning Shift · MCQ (Single Correct Answer)

If (1, 5, 35), (7, 5, 5), (1, λ, 7) and (2λ, 1, 2) are coplanar, then the sum of all possible values of λ is :

(A) $-\frac{44}{5}$ (B) $-\frac{39}{5}$ (C) $\frac{44}{5}$ (D) $\frac{39}{5}$

Open solution

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Q282. JEE Main 2021 (Online) 25th February Evening Shift · MCQ (Single Correct Answer)

A plane passes through the points A(1, 2, 3), B(2, 3, 1) and C(2, 4, 2). If O is the origin and P is (2, −1, 1), then the projection of $\overrightarrow{OP}$ on this plane is of length :

(A) $\sqrt{\frac{2}{7}}$ (B) $\sqrt{\frac{2}{5}}$ (C) $\sqrt{\frac{2}{3}}$ (D) $\sqrt{\frac{2}{11}}$

Open solution

Q283. JEE Main 2021 (Online) 24th February Evening Shift · MCQ (Single Correct Answer)

The vector equation of the plane passing through the intersection

of the planes $\vec{r}.\left(\hat{i}+\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(\hat{i}-2\hat{j}\right)=-2$, and the point (1, 0, 2) is :

(A) $\vec{r}.\left(\hat{i}+7\hat{j}+3\hat{k}\right)=\frac{7}{3}$

(B) $\vec{r}.\left(\hat{i}+7\hat{j}+3\hat{k}\right)=7$

(C) $\vec{r}.\left(3\hat{i}+7\hat{j}+3\hat{k}\right)=7$

(D) $\vec{r}.\left(\hat{i}-7\hat{j}+3\hat{k}\right)=\frac{7}{3}$

Open solution

Q284. JEE Main 2021 (Online) 24th February Morning Shift · MCQ (Single Correct Answer)

The equation of the plane passing through the point (1, 2, -3) and perpendicular to the planes

3x + y - 2z = 5 and 2x - 5y - z = 7, is :

(A) 6x - 5y + 2z + 10 =0 (B) 3x - 10y - 2z + 11 = 0

(C) 6x - 5y - 2z - 2 = 0 (D) 11x + y + 17z + 38 = 0

Open solution

Q285. JEE Main 2021 (Online) 24th February Morning Shift · MCQ (Single Correct Answer)

The distance of the point (1, 1, 9) from the point of intersection of the line $\frac{x-3}{1}=\frac{y-4}{2}=\frac{z-5}{2}$ and the plane x + y + z = 17 is :

(A) $19\sqrt{2}$ (B) $2\sqrt{19}$ (C) 38 (D) $\sqrt{38}$

Open solution

Q286. JEE Main 2020 (Online) 6th September Evening Slot · MCQ (Single Correct Answer)

A plane P meets the coordinate axes at A, B and C respectively. The centroid of ΔABC is given to be (1, 1, 2). Then the equation of the line through this centroid and perpendicular to the plane P is :

(A) $\frac{x-1}{1}=\frac{y-1}{1}=\frac{z-2}{2}$ (B) $\frac{x-1}{2}=\frac{y-1}{1}=\frac{z-2}{1}$

(C) $\frac{x-1}{2}=\frac{y-1}{2}=\frac{z-2}{1}$ (D) $\frac{x-1}{1}=\frac{y-1}{2}=\frac{z-2}{2}$

Open solution

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Q287. JEE Main 2020 (Online) 6th September Morning Slot · MCQ (Single Correct Answer)

The shortest distance between the lines

$\frac{x-1}{0} = \frac{y+1}{-1} = \frac{z}{1}$

and x + y + z + 1 = 0, 2x − y + z + 3 = 0 is :

(A) 1 (B) $\frac{1}{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{1}{\sqrt{3}}$

Open solution

Q288. JEE Main 2020 (Online) 5th September Evening Slot · MCQ (Single Correct Answer)

If for some $\alpha \in$ R, the lines

$L_1 : \frac{x+1}{2} = \frac{y-2}{-1} = \frac{z-1}{1}$ and

$L_2 : \frac{x+2}{\alpha} = \frac{y+1}{5-\alpha} = \frac{z+1}{1}$ are coplanar,

then the line L_2 passes through the point :

(A) (10, 2, 2) (B) (2, −10, −2) (C) (10, −2, −2) (D) (−2, 10, 2)

Open solution

Q289. JEE Main 2020 (Online) 4th September Evening Slot · MCQ (Single Correct Answer)

The distance of the point (1, −2, 3) from

the plane x − y + z = 5 measured parallel to

the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ is :

(A) 7 (B) 1 (C) $\frac{1}{7}$ (D) $\frac{7}{5}$

Open solution

Q290. JEE Main 2020 (Online) 3rd September Evening Slot · MCQ (Single Correct Answer)

The plane which bisects the line joining, the points (4, −2, 3) and (2, 4, −1) at right angles also passes through the point :

(A) (4, 0, 1) (B) (0, −1, 1) (C) (0, 1, −1) (D) (4, 0, −1)

Open solution

Q291. JEE Main 2020 (Online) 3rd September Morning Slot · MCQ (Single Correct Answer)

The foot of the perpendicular drawn from the point (4, 2, 3) to the line joining the points (1, −2, 3) and (1, 1, 0) lies on the plane :

(A) x − 2y + z = 1 (B) x + 2y − z = 1 (C) x − y − 2z = 1 (D) 2x + y − z = 1

Open solution

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Q292. JEE Main 2020 (Online) 2nd September Evening Slot · MCQ (Single Correct Answer)

A plane passing through the point (3, 1, 1) contains two lines whose direction ratios are 1, −2, 2 and 2, 3, −1 respectively. If this plane also passes through the point $(\alpha, -3, 5)$, then α is equal to:

(A) -10 (B) 10 (C) 5 (D) -5

Open solution

Q293. JEE Main 2020 (Online) 2nd September Morning Slot · MCQ (Single Correct Answer)

The plane passing through the points (1, 2, 1),
(2, 1, 2) and parallel to the line, 2x = 3y, z = 1
also passes through the point :

(A) (0, 6, −2) (B) (−2, 0, 1) (C) (0, −6, 2) (D) (2, 0 −1)

Open solution

Q294. JEE Main 2020 (Online) 8th January Evening Slot · MCQ (Single Correct Answer)

The mirror image of the point (1, 2, 3) in a plane is

$\left(-\frac{7}{3}, -\frac{4}{3}, -\frac{1}{3}\right)$. Which of the following points lies on this plane ?

(A) (1, −1, 1) (B) (−1, −1, −1) (C) (−1, −1, 1) (D) (1, 1, 1)

Open solution

Q295. JEE Main 2020 (Online) 7th January Morning Slot · MCQ (Single Correct Answer)

Let P be a plane passing through the points (2, 1, 0), (4, 1, 1) and (5, 0, 1) and R be any point (2, 1, 6). Then the image of R in the plane P is :

(A) (4, 3, 2) (B) (6, 5, - 2) (C) (3, 4, -2) (D) (6, 5, 2)

Open solution

Q296. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

A plane which bisects the angle between the two given planes 2x − y + 2z − 4 = 0 and x + 2y + 2z − 2 = 0, passes through the point :

(A) (1, −4, 1) (B) (1, 4, −1) (C) (2, 4, 1) (D) (2, −4, 1)

Open solution

Q297. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

The length of the perpendicular drawn from the point (2, 1, 4) to the plane containing the lines $\vec{r} = \left(\hat{i} + \hat{j}\right) + \lambda\left(\hat{i} + 2\hat{j} - \hat{k}\right)$ and $\vec{r} = \left(\hat{i} + \hat{j}\right) + \mu\left(-\hat{i} + \hat{j} - 2\hat{k}\right)$ is :

(A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) 3 (D) $\sqrt{3}$

Open solution

Q298. JEE Main 2019 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

If the line $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-1}{-1}$ intersects the plane 2x + 3y − z + 13 = 0 at a point P and the plane 3x + y + 4z = 16 at a point Q, then PQ is equal to :

(A) $2\sqrt{7}$ (B) 14 (C) $2\sqrt{14}$ (D) $\sqrt{14}$

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Q299. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

If the plane 2x − y + 2z + 3 = 0 has the distances $\frac{1}{3}$ and $\frac{2}{3}$ units from the planes 4x − 2y + 4z + λ = 0 and 2x − y + 2z + μ = 0, respectively, then the maximum value of $\lambda + \mu$ is equal to :

(A) 13 (B) 9 (C) 5 (D) 15

Open solution

Q300. JEE Main 2019 (Online) 10th April Evening Slot · MCQ (Single Correct Answer)

A perpendicular is drawn from a point on the line $\frac{x-1}{2} = \frac{y+1}{-1} = \frac{z}{1}$ to the plane x + y + z = 3 such that the foot of the perpendicular Q also lies on the plane x − y + z = 3. Then the co-ordinates of Q are :

(A) (4, 0, − 1) (B) (2, 0, 1) (C) (1, 0, 2) (D) (− 1, 0, 4)

Open solution

Q301. JEE Main 2019 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

If Q(0, −1, −3) is the image of the point P in the plane 3x − y + 4z = 2 and R is the point (3, −1, −2), then the area (in sq. units) of ΔPQR is :

(A) $\frac{\sqrt{65}}{2}$ (B) $2\sqrt{13}$ (C) $\frac{\sqrt{91}}{2}$ (D) $\frac{\sqrt{91}}{4}$

Open solution

Q302. JEE Main 2019 (Online) 9th April Evening Slot · MCQ (Single Correct Answer)

Let P be the plane, which contains the line of intersection of the planes, x + y + z − 6 = 0 and 2x + 3y + z + 5 = 0 and it is perpendicular to the xy-plane. Then the distance of the point (0, 0, 256) from P is equal to :

(A) $205\sqrt{5}$ (B) $63\sqrt{5}$ (C) $11/\sqrt{5}$ (D) $17/\sqrt{5}$

Open solution

Q303. JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

A plane passing through the points (0, −1, 0) and (0, 0, 1) and making an angle $\frac{\pi}{4}$ with the plane y − z + 5 = 0, also passes through the point

(A) $\left(\sqrt{2}, 1, 4\right)$ (B) $\left(-\sqrt{2}, 1, 4\right)$ (C) $\left(-\sqrt{2}, -1, -4\right)$ (D) $\left(\sqrt{2}, -1, 4\right)$

Open solution

Q304. JEE Main 2019 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If the line, $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{4}$ meets the plane, x + 2y + 3z = 15 at a point P, then the distance of P from the origin is :

(A) $\frac{\sqrt{5}}{2}$ (B) $2\sqrt{5}$ (C) 9/2 (D) 7/2

Open solution

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Q305. JEE Main 2019 (Online) 8th April Evening Slot · MCQ (Single Correct Answer)

The vector equation of the plane through the line of intersection of the planes x + y + z = 1 and 2x + 3y+ 4z = 5 which is perpendicular to the plane x − y + z = 0 is :

(A) $\vec{r} \times \left(\hat{i} - \hat{k}\right) - 2 = 0$

(B) $\vec{r}.\left(\hat{i} + \hat{k}\right) + 2 = 0$

(C) $\vec{r}.\left(\hat{i} - \hat{k}\right) + 2 = 0$

(D) $\vec{r} \times \left(\hat{i} - \hat{k}\right) + 2 = 0$

Open solution

Q306. JEE Main 2019 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The equation of a plane containing the line of intersection of the planes 2x − y − 4 = 0 and y + 2z − 4 = 0 and passing through the point (1, 1, 0) is :

(A) x − 3y − 2z = −2 (B) 2x − z = 2 (C) x − y − z = 0 (D) x + 3y + z = 4

Open solution

Q307. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

Let S be the set of all real values of λ such that a plane passing through the points $(-\lambda^2, 1, 1)$, $(1, -\lambda^2, 1)$ and $(1, 1, -\lambda^2)$ also passes through the point (−1, −1, 1). Then S is equal to :

(A) {1, −1} (B) {3, − 3}

(C) $\left\{\sqrt{3}\right\}$ (D) $\left\{\sqrt{3}, -\sqrt{3}\right\}$

Open solution

Q308. JEE Main 2019 (Online) 12th January Evening Slot · MCQ (Single Correct Answer)

If an angle between the line, $\frac{x+1}{2} = \frac{y-2}{1} = \frac{z-3}{-2}$ and the plane, $x - 2y - kz = 3$ is $\cos^{-1}\left(\frac{2\sqrt{2}}{3}\right)$, then a value of k is :

(A) $\sqrt{\frac{3}{5}}$ (B) $-\frac{5}{2}$ (C) $-\frac{3}{2}$ (D) $\sqrt{\frac{5}{3}}$

Open solution

Q309. JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

The perpendicular distance from the origin to the plane containing the two lines,

$\frac{x+2}{3} = \frac{y-2}{5} = \frac{z+5}{7}$ and

$\frac{x-1}{1} = \frac{y-4}{4} = \frac{z+4}{7}$, is :

(A) $6\sqrt{11}$ (B) $\frac{11}{\sqrt{6}}$ (C) 11 (D) $11\sqrt{6}$

Open solution

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Q310. JEE Main 2019 (Online) 12th January Morning Slot · MCQ (Single Correct Answer)

A tetrahedron has vertices P(1, 2, 1), Q(2, 1, 3), R(−1, 1, 2) and O(0, 0, 0). The angle between the faces OPQ and PQR is :

(A) $\cos^{-1}\left(\frac{17}{31}\right)$ (B) $\cos^{-1}\left(\frac{9}{35}\right)$

(C) $\cos^{-1}\left(\frac{19}{35}\right)$ (D) $\cos^{-1}\left(\frac{7}{31}\right)$

Open solution

Q311. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

Two lines $\frac{x-3}{1}=\frac{y+1}{3}=\frac{z-6}{-1}$ and $\frac{x+5}{7}=\frac{y-2}{-6}=\frac{z-3}{4}$ intersect at the point R. The reflection of R in the xy-plane has coordinates :

(A) (2, 4, 7) (B) (2, − 4, −7) (C) (2, − 4, 7) (D) (− 2, 4, 7)

Open solution

Q312. JEE Main 2019 (Online) 11th January Evening Slot · MCQ (Single Correct Answer)

If the point $(2, \alpha, \beta)$ lies on the plane which passes through the points (3, 4, 2) and (7, 0, 6) and is perpendicular to the plane 2x − 5y = 15, then $2\alpha - 3\beta$ is equal to

(A) 12 (B) 7 (C) 17 (D) 5

Open solution

Q313. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

The plane containing the line $\frac{x-3}{2}=\frac{y+2}{-1}=\frac{z-1}{3}$ and also containing its projection on the plane 2x + 3y − z = 5, contains which one of the following points ?

(A) (− 2, 2, 2) (B) (2, 2, 0) (C) (2, 0, − 2) (D) (0, − 2, 2)

Open solution

Q314. JEE Main 2019 (Online) 11th January Morning Slot · MCQ (Single Correct Answer)

The direction ratios of normal to the plane through the points (0, −1, 0) and (0, 0, 1) and making an angle $\frac{\pi}{4}$ with the plane y − z + 5 = 0 are :

(A) 2, −1, 1 (B) $2\sqrt{3}, 1, -1$ (C) $\sqrt{2}, 1, -1$ (D) $\sqrt{2}, -\sqrt{2}$

Open solution

Q315. JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

On which of the following lines lies the point of intersection of the line, $\frac{x-4}{2}=\frac{y-5}{2}=\frac{z-3}{1}$ and the plane, x + y + z = 2 ?

(A) $\frac{x-4}{1}=\frac{y-5}{1}=\frac{z-5}{-1}$ (B) $\frac{x-2}{2}=\frac{y-3}{2}=\frac{z+3}{3}$

(C) $\frac{x-1}{1}=\frac{y-3}{2}=\frac{z+4}{-5}$ (D) $\frac{x+3}{3}=\frac{4-y}{3}=\frac{z+1}{-2}$

Open solution

Q316. JEE Main 2019 (Online) 10th January Evening Slot · MCQ (Single Correct Answer)

The plane which bisects the line segment joining the points (−3, −3, 4) and (3, 7, 6) at right angles, passes through which one of the following points ?

(A) (2, 1, 3) (B) (4, − 1, 2) (C) (4, 1, − 2) (D) (− 2, 3, 5)

Open solution

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Q317. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

The plane passing through the point (4, −1, 2) and parallel to the lines $\frac{x+2}{3} = \frac{y-2}{-1} = \frac{z+1}{2}$ and $\frac{x-2}{1} = \frac{y-3}{2} = \frac{z-4}{3}$ also passes through the point -

(A) (1, 1, − 1) (B) (1, 1, 1) (C) (− 1, − 1, −1) (D) (− 1, − 1, 1)

Open solution

Q318. JEE Main 2019 (Online) 10th January Morning Slot · MCQ (Single Correct Answer)

Let A be a point on the line $\vec{r} = (1-3\mu)\hat{i} + (\mu-1)\hat{j} + (2+5\mu)\hat{k}$ and B(3, 2, 6) be a point in the space. Then the value of μ for which the vector $\overrightarrow{AB}$ is parallel to the plane x − 4y + 3z = 1 is -

(A) $\frac{1}{8}$ (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $-\frac{1}{4}$

Open solution

Q319. JEE Main 2019 (Online) 9th January Evening Slot · MCQ (Single Correct Answer)

The equation of the plane containing the straight line $\frac{x}{2} = \frac{y}{3} = \frac{z}{4}$ and perpendicular to the plane containing the straight lines $\frac{x}{3} = \frac{y}{4} = \frac{z}{2}$ and $\frac{x}{4} = \frac{y}{2} = \frac{z}{3}$ is :

(A) x − 2y + z = 0 (B) 3x + 2y − 3z = 0 (C) x + 2y − 2z = 0 (D) 5x + 2y − 4z = 0

Open solution

Q320. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

The equation of the line passing through (−4, 3, 1), parallel

to the plane x + 2y − z − 5 = 0 and intersecting

the line $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z-2}{-1}$ is :

(A) $\frac{x+4}{3} = \frac{y-3}{-1} = \frac{z-1}{1}$ (B) $\frac{x+4}{1} = \frac{y-3}{1} = \frac{z-1}{3}$

(C) $\frac{x+4}{-1} = \frac{y-3}{1} = \frac{z-1}{1}$ (D) $\frac{x-4}{2} = \frac{y+3}{1} = \frac{z+1}{4}$

Open solution

Q321. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

The plane through the intersection of the planes x + y + z = 1 and 2x + 3y − z + 4 = 0 and parallel to y-axis also passes through the point :

(A) (−3, 0, -1) (B) (3, 2, 1) (C) (3, 3, -1) (D) (−3, 1, 1)

Open solution

Q322. JEE Main 2018 (Online) 16th April Morning Slot · MCQ (Single Correct Answer)

The sum of the intercepts on the coordinate axes of the plane passing through the point $(-2, -2, 2)$ and containing the line joining the points $(1, -1, 2)$ and (1, 1, 1) is :

(A) 4 (B) − 4 (C) − 8 (D) 12

Open solution

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Q323. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

If L_1 is the line of intersection of the planes 2x - 2y + 3z - 2 = 0, x - y + z + 1 = 0 and L_2 is the line of intersection of the planes x + 2y - z - 3 = 0, 3x - y + 2z - 1 = 0, then the distance of the origin from the plane, containing the lines L_1 and L_2, is :

Open solution

(A) $\frac{1}{\sqrt{2}}$ (B) $\frac{1}{4\sqrt{2}}$ (C) $\frac{1}{3\sqrt{2}}$ (D) $\frac{1}{2\sqrt{2}}$

Q324. JEE Main 2018 (Offline) · MCQ (Single Correct Answer)

The length of the projection of the line segment joining the points (5, -1, 4) and (4, -1, 3) on the plane, x + y + z = 7 is :

Open solution

(A) $\sqrt{\frac{2}{3}}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{2}{3}$ (D) $\frac{1}{3}$

Q325. JEE Main 2018 (Online) 15th April Evening Slot · MCQ (Single Correct Answer)

A plane bisects the line segment joining the points (1, 2, 3) and $(-3, 4, 5)$ at rigt angles. Then this plane also passes through the point :

Open solution

(A) $(-3, 2, 1)$ (B) (3, 2, 1) (C) $(-1, 2, 3)$ (D) $(1, 2, -3)$

Q326. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

A variable plane passes through a fixed point (3,2,1) and meets x, y and z axes at A, B and C respectively. A plane is drawn parallel to yz -plane through A, a second plane is drawn parallel zx-plane through B and a third plane is drawn parallel to xy-plane through C. Then the locus of the point of intersection of these three planes, is :

Open solution

(A) $\frac{x}{3}+\frac{y}{2}+\frac{z}{1}=1$ (B) x + y + z = 6

(C) $\frac{1}{x}+\frac{1}{y}+\frac{1}{z}=\frac{11}{6}$ (D) $\frac{3}{x}+\frac{2}{y}+\frac{1}{z}=1$

Q327. JEE Main 2018 (Online) 15th April Morning Slot · MCQ (Single Correct Answer)

An angle between the plane, x + y + z = 5 and the line of intersection of the planes, 3x + 4y + z $-$ 1 = 0 and 5x + 8y + 2z + 14 =0, is :

Open solution

(A) $\sin^{-1}\left(\sqrt{3/17}\right)$

(B) $\cos^{-1}\left(\sqrt{3/17}\right)$

(C) $\cos^{-1}(3/17)$

(D) $\sin^{-1}(3/17)$

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Q328. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If a variable plane, at a distance of 3 units from the origin, intersects the coordinate axes at A, B and C, then the locus of the centroid of ΔABC is :

(A) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=1$ (B) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=3$

(C) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{9}$ (D) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=9$

Open solution

Q329. JEE Main 2017 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

If the line, $\frac{x-3}{1}=\frac{y+2}{-1}=\frac{z+\lambda}{-2}$ lies in the plane, 2x−4y+3z=2, then the shortest distance between this line and the line, $\frac{x-1}{12}=\frac{y}{9}=\frac{z}{4}$ is :

(A) 2 (B) 1 (C) 0 (D) 3

Open solution

Q330. JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The coordinates of the foot of the perpendicular from the point (1, −2, 1) on the plane containing the lines, $\frac{x+1}{6}=\frac{y-1}{7}=\frac{z-3}{8}$ and $\frac{x-1}{3}=\frac{y-2}{5}=\frac{z-3}{7}$, is :

(A) (2, −4, 2) (B) (− 1, 2, −1) (C) (0, 0, 0) (D) (1, 1, 1)

Open solution

Q331. JEE Main 2017 (Online) 8th April Morning Slot · MCQ (Single Correct Answer)

The line of intersection of the planes $\vec{r}.\left(3\hat{i}-\hat{j}+\hat{k}\right)=1$ and $\vec{r}.\left(\hat{i}+4\hat{j}-2\hat{k}\right)=2$, is :

(A) $\frac{x-\frac{4}{7}}{-2}=\frac{y}{7}=\frac{z-\frac{5}{7}}{13}$

(B) $\frac{x-\frac{4}{7}}{2}=\frac{y}{-7}=\frac{z+\frac{5}{7}}{13}$

(C) $\frac{x-\frac{6}{13}}{2}=\frac{y-\frac{5}{13}}{-7}=\frac{z}{-13}$

(D) $\frac{x-\frac{6}{13}}{2}=\frac{y-\frac{5}{13}}{7}=\frac{z}{-13}$

Open solution

Q332. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

The distance of the point (1, 3, − 7) from the plane passing through the point (1, −1, − 1), having normal perpendicular to both the lines

$\frac{x-1}{1}=\frac{y+2}{-2}=\frac{z-4}{3}$

and

$\frac{x-2}{2}=\frac{y+1}{-1}=\frac{z+7}{-1}$ is :

(A) $\frac{10}{\sqrt{83}}$ (B) $\frac{5}{\sqrt{83}}$ (C) $\frac{10}{\sqrt{74}}$ (D) $\frac{20}{\sqrt{74}}$

Open solution

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Q333. JEE Main 2017 (Offline) · MCQ (Single Correct Answer)

If the image of the point P(1, −2, 3) in the plane, 2x + 3y − 4z + 22 = 0 measured parallel to the line,

$\frac{x}{1} = \frac{y}{4} = \frac{z}{5}$ is Q, then PQ is equal to:

(A) $2\sqrt{42}$ (B) $\sqrt{42}$ (C) $6\sqrt{5}$ (D) $3\sqrt{5}$

Open solution

Q334. JEE Main 2016 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

The number of distinct real values of λ for which the lines

$\frac{x-1}{1} = \frac{y-2}{2} = \frac{z+3}{\lambda^2}$ and $\frac{x-3}{1} = \frac{y-2}{\lambda^2} = \frac{z-1}{2}$ are coplanar is :

(A) 4 (B) 1 (C) 2 (D) 3

Open solution

Q335. JEE Main 2016 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

The distance of the point (1, − 2, 4) from the plane passing through the point (1, 2, 2) and perpendicular to the planes x − y + 2z = 3 and 2x − 2y + z + 12 = 0, is :

(A) $2\sqrt{2}$ (B) 2 (C) $\sqrt{2}$ (D) $\frac{1}{\sqrt{2}}$

Open solution

Q336. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

If the line, $\frac{x-3}{2} = \frac{y+2}{-1} = \frac{z+4}{3}$ lies in the planes, $lx + my - z = 9$, then $l^2 + m^2$ is equal to :

(A) 5 (B) 2 (C) 26 (D) 18

Open solution

Q337. JEE Main 2016 (Offline) · MCQ (Single Correct Answer)

The distance of the point $(1, -5, 9)$ from the plane $x - y + z = 5$ measured along the line $x = y = z$ is :

(A) $\frac{10}{\sqrt{3}}$ (B) $\frac{20}{3}$ (C) $3\sqrt{10}$ (D) $10\sqrt{3}$

Open solution

Q338. JEE Main 2015 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

A plane containing the point (3, 2, 0) and the line

$\frac{x-1}{1} = \frac{y-2}{5} = \frac{z-3}{4}$ also contains the point :

(A) (0, −3, 1) (B) (0, 7, 10) (C) (0, 7, −10) (D) (0, 3, 1)

Open solution

Q339. JEE Main 2015 (Online) 10th April Morning Slot · MCQ (Single Correct Answer)

If the points (1, 1, λ) and (−3, 0, 1) are equidistant from the plane, 3x + 4y − 12z + 13 = 0, then λ satisfies the equation :

(A) $3x^2 - 10x + 7 = 0$ (B) $3x^2 + 10x + 7 = 0$

(C) $3x^2 + 10x - 13 = 0$ (D) $3x^2 - 10x + 21 = 0$

Open solution

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Q340. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

The equation of the plane containing the line $2x - 5y + z = 3; x + y + 4z = 5$, and parallel to the plane, $x + 3y + 6z = 1$, is :

(A) $x + 3y + 6z = 7$ (B) $2x + 6y + 12z = -13$

(C) $2x + 6y + 12z = 13$ (D) $x + 3y + 6z = -7$

Open solution

Q341. JEE Main 2015 (Offline) · MCQ (Single Correct Answer)

The distance of the point $(1, 0, 2)$ from the point of intersection of the line $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12}$ and the plane $x - y + z = 16$, is :

(A) $3\sqrt{21}$ (B) 13 (C) $2\sqrt{14}$ (D) 8

Open solution

Q342. JEE Main 2014 (Online) 19th April Morning Slot · MCQ (Single Correct Answer)

If the angle between the line 2(x + 1) = y = z + 4 and
the plane 2x $-$ y + $\sqrt{\lambda}$ z + 4 = 0 is $\frac{\pi}{6}$, then the value of λ is :

(A) $\frac{135}{7}$ (B) $\frac{45}{11}$ (C) $\frac{45}{7}$ (D) $\frac{135}{11}$

Open solution

Q343. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

A symmetrical form of the line of intersection of the planes
x = ay + b and z = cy + d is :

(A) $\frac{x-b}{a} = \frac{y-1}{1} = \frac{z-d}{c}$ (B) $\frac{x-b-a}{a} = \frac{y-1}{1} = \frac{z-d-c}{c}$

(C) $\frac{x-a}{b} = \frac{y-0}{1} = \frac{z-c}{d}$ (D) $\frac{x-b-a}{b} = \frac{y-1}{0} = \frac{z-d-c}{d}$

Open solution

Q344. JEE Main 2014 (Online) 12th April Morning Slot · MCQ (Single Correct Answer)

If the distance between planes, 4x $-$ 2y $-$ 4z + 1 = 0 and
4x $-$ 2y $-$ 4z + d = 0 is 7, then d is :

(A) 41 or $-$ 42 (B) 42 or $-$ 43 (C) $-$ 41 or 43 (D) $-$ 42 or 44

Open solution

Q345. JEE Main 2014 (Online) 11th April Morning Slot · MCQ (Single Correct Answer)

The plane containing the line $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and parallel to the line $\frac{x}{1} = \frac{y}{1} = \frac{z}{4}$ passes through the point :

(A) (1, $-$ 2, 5) (B) (1, 0, 5) (C) (0, 3, $-$5) (D) ($-$ 1, $-$ 3, 0)

Open solution

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Q346. JEE Main 2014 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

Equation of the plane which passes through the point of intersection of lines

$\frac{x-1}{3} = \frac{y-2}{1} = \frac{z-3}{2}$ and $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z-2}{3}$

and has the largest distance from the origin is :

(A) 7x + 2y + 4z = 54
(B) 3x + 4y + 5z = 49
(C) 4x + 3y + 5z = 50
(D) 5x + 4y + 3z = 57

Open solution

Q347. JEE Main 2014 (Offline) · MCQ (Single Correct Answer)

The image of the line $\frac{x-1}{3} = \frac{y-3}{1} = \frac{z-4}{-5}$ in the plane $2x - y + z + 3 = 0$ is the line :

(A) $\frac{x-3}{3} = \frac{y+5}{1} = \frac{z-2}{-5}$
(B) $\frac{x-3}{-3} = \frac{y+5}{-1} = \frac{z-2}{5}$
(C) $\frac{x+3}{3} = \frac{y-5}{1} = \frac{z-2}{-5}$
(D) $\frac{x+3}{-3} = \frac{y-5}{-1} = \frac{z+2}{5}$

Open solution

Q348. JEE Main 2013 (Online) 25th April Morning Slot · MCQ (Single Correct Answer)

The equation of a plane through the line of intersection of the planes x + 2y = 3, y − 2z + 1 = 0, and perpendicular to the first plane is :

(A) 2x − y − 10z = 9
(B) 2x − y + 7z = 11
(C) 2x − y + 10z = 11
(D) 2x − y − 9z = 10

Open solution

Q349. JEE Main 2013 (Online) 22th April Morning Slot · MCQ (Single Correct Answer)

Let Q be the foot of perpendicular from the origin to the plane 4x − 3y + z + 13 = 0 and R be a point (− 1, 1, − 6) on the plane. Then length QR is :

(A) $\sqrt{14}$
(B) $\sqrt{\frac{19}{2}}$
(C) $3\sqrt{\frac{7}{2}}$
(D) $\frac{3}{\sqrt{2}}$

Open solution

Q350. JEE Main 2013 (Online) 9th April Morning Slot · MCQ (Single Correct Answer)

A vector $\vec{n}$ is inclined to x-axis at 45°, to y-axis at 60° and at an acute angle to z-axis. If $\vec{n}$ is a normal to a plane passing through the point $(\sqrt{2}, -1, 1)$ then the equation of the plane is :

(A) $4\sqrt{2}$x + 7y + z = 2
(B) 2x + y + 2z = $2\sqrt{2}$ + 1
(C) $3\sqrt{2}$x − 4y − 3z = 7
(D) $\sqrt{2}$x + y + z = 2

Open solution

Q351. JEE Main 2013 (Offline) · MCQ (Single Correct Answer)

Distance between two parallel planes $2x + y + 2z = 8$ and $4x + 2y + 4z + 5 = 0$ is :

(A) $\frac{3}{2}$
(B) $\frac{5}{2}$
(C) $\frac{7}{2}$
(D) $\frac{9}{2}$

Open solution

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Q352. AIEEE 2012 (Online) 26th May · MCQ (Single Correct Answer)

Open solution

Consider the following planes
$P : x + y - 2z + 7 = 0$
$Q : x + y + 2z + 2 = 0$
$R : 3x + 3y - 6z - 11 = 0$

(A) P and R are perpendicular
(B) Q and R are perpendicular
(C) P and Q are parallel
(D) P and R are parallel

Q353. AIEEE 2012 (Online) 26th May · MCQ (Single Correct Answer)

Open solution

The equation of a plane containing the line $\frac{x+1}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$ and the point $(0, 7, -7)$ is

(A) $x + y + z = 0$
(B) $x + 2y + z = 21$
(C) $3x - 2y + 5z + 35 = 0$
(D) $3x + 2y + 5z + 21 = 0$

Q354. AIEEE 2012 (Online) 19th May · MCQ (Single Correct Answer)

Open solution

If the three planes $x = 5$, $2x - 5ay + 3z - 2 = 0$ and $3bx + y - 3z = 0$ contain a common line, then (a, b) is equal to

(A) $\left(\frac{8}{15}, -\frac{1}{5}\right)$
(B) $\left(\frac{1}{5}, -\frac{8}{15}\right)$
(C) $\left(-\frac{8}{15}, \frac{1}{5}\right)$
(D) $\left(-\frac{1}{5}, \frac{8}{15}\right)$

Q355. AIEEE 2012 (Online) 7th May · MCQ (Single Correct Answer)

Open solution

The values of a for which the two points $(1, a, 1)$ and $(-3, 0, a)$ lie on the opposite sides of the plane $3x + 4y - 12z + 13 = 0$, satisfy

(A) $0 < a < \frac{1}{3}$
(B) $-1 < a < 0$
(C) $a < -1$ or $a > \frac{1}{3}$
(D) $a = 0$

Q356. AIEEE 2012 (Online) 7th May · MCQ (Single Correct Answer)

Open solution

A line with positive direction cosines passes through the point $P(2, -1, 2)$ and makes equal angles with the coordinate axes. If the line meets the plane $2x + y + z = 9$ at point Q, then the length PQ equals

(A) $\sqrt{2}$
(B) 2
(C) $\sqrt{3}$
(D) 1

Q357. AIEEE 2012 · MCQ (Single Correct Answer)

Open solution

A equation of a plane parallel to the plane $x - 2y + 2z - 5 = 0$ and at a unit distance from the origin is :

(A) $x - 2y + 2z - 3 = 0$
(B) $x - 2y + 2z + 1 = 0$
(C) $x - 2y + 2z - 1 = 0$
(D) $x - 2y + 2z + 5 = 0$

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Q358. AIEEE 2011 · MCQ (Single Correct Answer)

If the angle between the line $x = \frac{y-1}{2} = \frac{z-3}{\lambda}$ and the plane

$x + 2y + 3z = 4$ is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then λ equals :

(A) $\frac{3}{2}$ (B) $\frac{2}{5}$ (C) $\frac{5}{3}$ (D) $\frac{2}{3}$

Open solution

Q359. AIEEE 2010 · MCQ (Single Correct Answer)

Statement-1 : The point $A(3, 1, 6)$ is the mirror image of the point $B(1, 3, 4)$ in the plane $x - y + z = 5$.

Statement-2 : The plane $x - y + z = 5$ bisects the line segment joining $A(3, 1, 6)$ and $B(1, 3, 4)$.

(A) Statement - 1 is true, Statement - 2 is true; Statement - 2 is **not** a correct explanation for Statement - 1.

(B) Statement - 1 is true, Statement - 2 is false.

(C) Statement - 1 is false , Statement - 2 is true.

(D) Statement - 1 is true, Statement - 2 is true; Statement - 2 is a correct explanation for Statement - 1.

Open solution

Q360. AIEEE 2009 · MCQ (Single Correct Answer)

Let the line $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ lie in the plane $x + 3y - \alpha z + \beta = 0$. Then (α, β) equals

(A) $(-6, 7)$ (B) $(5, -15)$ (C) $(-5, 5)$ (D) $(6, -17)$

Open solution

Q361. AIEEE 2007 · MCQ (Single Correct Answer)

If $(2, 3, 5)$ is one end of a diameter of the sphere $x^2 + y^2 + z^2 - 6x - 12y - 2z + 20 = 0$, then the coordinates of the other end of the diameter are

(A) $(4, 3, 5)$ (B) $(4, 3, -3)$ (C) $(4, 9, -3)$ (D) $(4, -3, 3)$

Open solution

Q362. AIEEE 2006 · MCQ (Single Correct Answer)

The image of the point $(-1, 3, 4)$ in the plane $x - 2y = 0$ is :

(A) $\left(-\frac{17}{3}, -\frac{19}{3}, 4\right)$ (B) $(15, 11, 4)$

(C) $\left(-\frac{17}{3}, -\frac{19}{3}, 1\right)$ (D) None of these

Open solution

Q363. AIEEE 2005 · MCQ (Single Correct Answer)

If the angle θ between the line $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z-2}{2}$ and

the plane $2x - y + \sqrt{\lambda}\, z + 4 = 0$ is such that $\sin\ \theta = \frac{1}{3}$ then value of λ is :

(A) $\frac{5}{3}$ (B) $\frac{-3}{5}$ (C) $\frac{3}{4}$ (D) $\frac{-4}{3}$

Open solution

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Q364. AIEEE 2005 · MCQ (Single Correct Answer)

If the plane $2ax - 3ay + 4az + 6 = 0$ passes through the midpoint of the line joining the centres of the spheres

$x^2 + y^2 + z^2 + 6x - 8y - 2z = 13$ and

$x^2 + y^2 + z^2 - 10x + 4y - 2z = 8$ then a equals :

(A) -1 (B) 1 (C) -2 (D) 2

Open solution

Q365. AIEEE 2005 · MCQ (Single Correct Answer)

The distance between the line

$\overrightarrow{r} = 2\hat{i} - 2\hat{j} + 3\widehat{k} + \lambda\,(i - j + 4k)$, and the plane

$\overrightarrow{r}.\left(\hat{i} + 5\hat{j} + \widehat{k}\right) = 5$ is

(A) $\frac{10}{9}$ (B) $\frac{10}{3\sqrt{3}}$ (C) $\frac{3}{10}$ (D) $\frac{10}{3}$

Open solution

Q366. AIEEE 2005 · MCQ (Single Correct Answer)

The plane $x + 2y - z = 4$ cuts the sphere $x^2 + y^2 + z^2 - x + z - 2 = 0$ in a circle of radius

(A) 3 (B) 1 (C) 2 (D) $\sqrt{2}$

Open solution

Q367. AIEEE 2004 · MCQ (Single Correct Answer)

Distance between two parallel planes

$2x + y + 2z = 8$ and $4x + 2y + 4z + 5 = 0$ is :

(A) $\frac{9}{2}$ (B) $\frac{5}{2}$ (C) $\frac{7}{2}$ (D) $\frac{3}{2}$

Open solution

Q368. AIEEE 2004 · MCQ (Single Correct Answer)

The intersection of the spheres
$x^2 + y^2 + z^2 + 7x - 2y - z = 13$ and
$x^2 + y^2 + z^2 - 3x + 3y + 4z = 8$
is the same as the intersection of one of the sphere and the plane

(A) $2x - y - z = 1$ (B) $x - 2y - z = 1$ (C) $x - y - 2z = 1$ (D) $x - y - z = 1$

Open solution

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Q369. AIEEE 2003 · MCQ (Single Correct Answer)

The shortest distance from the plane $12x + 4y + 3z = 327$ to the sphere

$x^2 + y^2 + z^2 + 4x - 2y - 6z = 155$ is

(A) 39 (B) 26 (C) $11\frac{4}{13}$ (D) 13

Open solution

Q370. AIEEE 2003 · MCQ (Single Correct Answer)

The radius of the circle in which the sphere

$x^2 + y^2 + z^2 + 2x - 2y - 4z - 19 = 0$ is cut by the plane

$x + 2y + 2z + 7 = 0$ is

(A) 4 (B) 1 (C) 2 (D) 3

Open solution

Q371. AIEEE 2003 · MCQ (Single Correct Answer)

Two systems of rectangular axes have the same origin. If a plane cuts then at distances a, b, c and a', b', c' from the origin then

(A) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} - \frac{1}{a'^2} - \frac{1}{b'^2} - \frac{1}{c'^2} = 0$

(B) $\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} + \frac{1}{a'^2} + \frac{1}{b'^2} + \frac{1}{c'^2} = 0$

(C) $\frac{1}{a^2} + \frac{1}{b^2} - \frac{1}{c^2} + \frac{1}{a'^2} - \frac{1}{b'^2} - \frac{1}{c'^2} = 0$

(D) $\frac{1}{a^2} - \frac{1}{b^2} - \frac{1}{c^2} + \frac{1}{a'^2} - \frac{1}{b'^2} - \frac{1}{c'^2} = 0$

Open solution

Q372. AIEEE 2002 · MCQ (Single Correct Answer)

A plane which passes through the point $(3, 2, 0)$ and the line

$\frac{x-4}{1} = \frac{y-7}{5} = \frac{z-4}{4}$ is :

(A) $x - y + z = 1$ (B) $x + y + z = 5$ (C) $x + 2y - z = 1$ (D) $2x - y + z = 5$

Open solution

Q373. AIEEE 2002 · MCQ (Single Correct Answer)

The $d.r.$ of normal to the plane through $(1, 0, 0), (0, 1, 0)$ which makes an angle $\pi/4$ with plane $x + y = 3$ are :

(A) $1, \sqrt{2}, 1$ (B) $1, 1, \sqrt{2}$ (C) $1, 1, 2$ (D) $\sqrt{2}, 1, 1$

Open solution

Q374. JEE Main 2023 (Online) 15th April Morning Shift · Numerical

Let the plane P contain the line $2x + y - z - 3 = 0 = 5x - 3y + 4z + 9$ and be parallel to the line $\frac{x+2}{2} = \frac{3-y}{-4} = \frac{z-7}{5}$. Then the distance of the point $\mathrm{A}(8, -1, -19)$ from the plane P measured parallel to the line $\frac{x}{-3} = \frac{y-5}{4} = \frac{2-z}{-12}$ is equal to _____.

Open solution

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Q375. JEE Main 2023 (Online) 13th April Morning Shift · Numerical

Let the image of the point $\left(\frac{5}{3}, \frac{5}{3}, \frac{8}{3}\right)$ in the plane $x - 2y + z - 2 = 0$ be P. If the distance of the point $Q(6, -2, \alpha), \alpha > 0$, from P is 13 , then α is equal to _____.

Open solution

Q376. JEE Main 2023 (Online) 12th April Morning Shift · Numerical

Let the plane $x + 3y - 2z + 6 = 0$ meet the co-ordinate axes at the points A, B, C. If the orthocenter of the triangle ABC is $\left(\alpha, \beta, \frac{6}{7}\right)$, then $98(\alpha + \beta)^2$ is equal to _____.

Open solution

Q377. JEE Main 2023 (Online) 11th April Evening Shift · Numerical

Let the line $l : x = \frac{1-y}{-2} = \frac{z-3}{\lambda}, \lambda \in \mathbb{R}$ meet the plane $P : x + 2y + 3z = 4$ at the point (α, β, γ). If the angle between the line l and the plane P is $\cos^{-1}\left(\sqrt{\frac{5}{14}}\right)$, then $\alpha + 2\beta + 6\gamma$ is equal to _____.

Open solution

Q378. JEE Main 2023 (Online) 10th April Evening Shift · Numerical

Let the foot of perpendicular from the point $\mathrm{A}(4, 3, 1)$ on the plane $\mathrm{P} : x - y + 2z + 3 = 0$ be N. If $\mathrm{B}(5, \alpha, \beta), \alpha, \beta \in \mathbb{Z}$ is a point on plane P such that the area of the triangle ABN is $3\sqrt{2}$, then $\alpha^2 + \beta^2 + \alpha\beta$ is equal to _____.

Open solution

Q379. JEE Main 2023 (Online) 8th April Evening Shift · Numerical

Let P_1 be the plane $3x - y - 7z = 11$ and P_2 be the plane passing through the points $(2, -1, 0), (2, 0, -1)$, and $(5, 1, 1)$. If the foot of the perpendicular drawn from the point $(7, 4, -1)$ on the line of intersection of the planes P_1 and P_2 is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to _____.

Open solution

Q380. JEE Main 2023 (Online) 8th April Morning Shift · Numerical

Let λ_1, λ_2 be the values of λ for which the points $\left(\frac{5}{2}, 1, \lambda\right)$ and $(-2, 0, 1)$ are at equal distance from the plane $2x + 3y - 6z + 7 = 0$. If $\lambda_1 > \lambda_2$, then the distance of the point $(\lambda_1 - \lambda_2, \lambda_2, \lambda_1)$ from the line $\frac{x-5}{1} = \frac{y-1}{2} = \frac{z+7}{2}$ is _____.

Open solution

Q381. JEE Main 2023 (Online) 6th April Morning Shift · Numerical

Let the image of the point $\mathrm{P}(1, 2, 3)$ in the plane $2x - y + z = 9$ be Q. If the coordinates of the point R are $(6, 10, 7)$, then the square of the area of the triangle PQR is _____.

Open solution

Q382. JEE Main 2023 (Online) 1st February Evening Shift · Numerical

The point of intersection C of the plane $8x + y + 2z = 0$ and the line joining the points $\mathrm{A}(-3, -6, 1)$ and $\mathrm{B}(2, 4, -3)$ divides the line segment AB internally in the ratio $\mathrm{k} : 1$. If $\mathrm{a}, \mathrm{b}, \mathrm{c}(|\mathrm{a}|, |\mathrm{b}|, |\mathrm{c}|$ are coprime) are the direction ratios of the perpendicular from the point C on the line $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$, then $|\mathrm{a} + \mathrm{b} + \mathrm{c}|$ is equal to _____.

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Q383. JEE Main 2023 (Online) 1st February Evening Shift · Numerical

Let $\alpha x + \beta y + \gamma z = 1$ be the equation of a plane passing through the point $(3, -2, 5)$ and perpendicular to the line joining the points $(1, 2, 3)$ and $(-2, 3, 5)$. Then the value of $\alpha\beta\gamma$ is equal to _____.

Open solution

Q384. JEE Main 2023 (Online) 31st January Morning Shift · Numerical

Let the line $L : \frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{1}$ intersect the plane $2x + y + 3z = 16$ at the point P. Let the point Q be the foot of perpendicular from the point $R(1, -1, -3)$ on the line L. If α is the area of triangle PQR, then α^2 is equal to _____.

Open solution

Q385. JEE Main 2023 (Online) 31st January Morning Shift · Numerical

Let θ be the angle between the planes $P_1 : \vec{r} \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 9$ and $P_2 : \vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 15$. Let L be the line that meets P_2 at the point $(4, -2, 5)$ and makes an angle θ with the normal of P_2. If α is the angle between L and P_2, then $(\tan^2\theta)\,(\cot^2\alpha)$ is equal to _____.

Open solution

Q386. JEE Main 2023 (Online) 30th January Morning Shift · Numerical

If the equation of the plane passing through the point $(1, 1, 2)$ and perpendicular to the line $x - 3y + 2z - 1 = 0 = 4x - y + z$ is $\mathrm{A}x + \mathrm{B}y + \mathrm{C}z = 1$, then $140(\mathrm{C} - \mathrm{B} + \mathrm{A})$ is equal to _____.

Open solution

Q387. JEE Main 2023 (Online) 30th January Morning Shift · Numerical

If $\lambda_1 < \lambda_2$ are two values of λ such that the angle between the planes $P_1 : \vec{r}(3\hat{i} - 5\hat{j} + \hat{k}) = 7$ and $P_2 : \vec{r} \cdot (\lambda\hat{i} + \hat{j} - 3\hat{k}) = 9$ is $\sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$, then the square of the length of perpendicular from the point $(38\lambda_1, 10\lambda_2, 2)$ to the plane P_1 is _____.

Open solution

Q388. JEE Main 2023 (Online) 29th January Morning Shift · Numerical

Let the equation of the plane P containing the line $x + 10 = \frac{8-y}{2} = z$ be $ax + by + 3z = 2(a + b)$ and the distance of the plane P from the point (1, 27, 7) be c. Then $a^2 + b^2 + c^2$ is equal to _____.

Open solution

Q389. JEE Main 2023 (Online) 25th January Morning Shift · Numerical

Let the equation of the plane passing through the line $x - 2y - z - 5 = 0 = x + y + 3z - 5$ and parallel to the line $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ be $ax + by + cz = 65$. Then the distance of the point (a, b, c) from the plane $2x + 2y - z + 16 = 0$ is _____.

Open solution

Q390. JEE Main 2022 (Online) 29th July Morning Shift · Numerical

Let a line with direction ratios $a, -4a, -7$ be perpendicular to the lines with direction ratios $3, -1, 2b$ and $b, a, -2$. If the point of intersection of the line $\frac{x+1}{a^2+b^2} = \frac{y-2}{a^2-b^2} = \frac{z}{1}$ and the plane $x - y + z = 0$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to _____.

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Q391. JEE Main 2022 (Online) 27th July Morning Shift · **Numerical**

Let the line $\frac{x-3}{7}=\frac{y-2}{-1}=\frac{z-3}{-4}$ intersect the plane containing the lines $\frac{x-4}{1}=\frac{y+1}{-2}=\frac{z}{1}$ and $4ax-y+5z-7a=0=2x-5y-z-3, a\in\mathbb{R}$ at the point $P(\alpha,\beta,\gamma)$. Then the value of $\alpha+\beta+\gamma$ equals _____.

Open solution

Q392. JEE Main 2022 (Online) 26th July Evening Shift · **Numerical**

The largest value of a, for which the perpendicular distance of the plane containing the lines $\vec{r}=(\hat{i}+\hat{j})+\lambda(\hat{i}+a\hat{j}-\hat{k})$ and $\vec{r}=(\hat{i}+\hat{j})+\mu(-\hat{i}+\hat{j}-a\hat{k})$ from the point $(2,1,4)$ is $\sqrt{3}$, is _____.

Open solution

Q393. JEE Main 2022 (Online) 26th July Evening Shift · **Numerical**

The plane passing through the line $L: lx-y+3(1-l)z=1, x+2y-z=2$ and perpendicular to the plane $3x+2y+z=6$ is $3x-8y+7z=4$. If θ is the acute angle between the line L and the y-axis, then $415\cos^2\theta$ is equal to _____.

Open solution

Q394. JEE Main 2022 (Online) 25th July Morning Shift · **Numerical**

The line of shortest distance between the lines $\frac{x-2}{0}=\frac{y-1}{1}=\frac{z}{1}$ and $\frac{x-3}{2}=\frac{y-5}{2}=\frac{z-1}{1}$ makes an angle of $\cos^{-1}\left(\sqrt{\frac{2}{27}}\right)$ with the plane $\mathrm{P}: \mathrm{a}x-y-z=0, (a>0)$. If the image of the point $(1,1,-5)$ in the plane P is (α,β,γ), then $\alpha+\beta-\gamma$ is equal to _____.

Open solution

Q395. JEE Main 2022 (Online) 29th June Morning Shift · **Numerical**

Let d be the distance between the foot of perpendiculars of the points P(1, 2, −1) and Q(2, −1, 3) on the plane −x + y + z = 1. Then d^2 is equal to _____.

Open solution

Q396. JEE Main 2022 (Online) 29th June Morning Shift · **Numerical**

Let $P_1: \vec{r}.\left(2\hat{i}+\hat{j}-3\hat{k}\right)=4$ be a plane. Let P_2 be another plane which passes through the points (2, −3, 2), (2, −2, −3) and (1, −4, 2). If the direction ratios of the line of intersection of P_1 and P_2 be 16, α, β, then the value of $\alpha+\beta$ is equal to _____.

Open solution

Q397. JEE Main 2022 (Online) 27th June Morning Shift · **Numerical**

Let the mirror image of the point (a, b, c) with respect to the plane 3x − 4y + 12z + 19 = 0 be (a − 6, β, γ). If a + b + c = 5, then $7\beta-9\gamma$ is equal to _____.

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Q398. JEE Main 2022 (Online) 25th June Morning Shift · Numerical

Let the lines

$L_1 : \vec{r} = \lambda\left(\hat{i} + 2\hat{j} + 3\hat{k}\right),\ \lambda \in R$

$L_2 : \vec{r} = \left(\hat{i} + 3\hat{j} + \hat{k}\right) + \mu\left(\hat{i} + \hat{j} + 5\hat{k}\right);\ \mu \in R,$

intersect at the point S. If a plane ax + by − z + d = 0 passes through S and is parallel to both the lines L_1 and L_2, then the value of a + b + d is equal to _____.

Open solution

Q399. JEE Main 2021 (Online) 31st August Evening Shift · Numerical

Suppose, the line $\frac{x-2}{\alpha} = \frac{y-2}{-5} = \frac{z+2}{2}$ lies on the plane $x + 3y - 2z + \beta = 0$. Then $(\alpha + \beta)$ is equal to _____.

Open solution

Q400. JEE Main 2021 (Online) 31st August Morning Shift · Numerical

The square of the distance of the point of intersection

of the line $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z+1}{6}$ and the plane $2x - y + z = 6$ from the point $(-1, -1, 2)$ is _____.

Open solution

Q401. JEE Main 2021 (Online) 27th August Evening Shift · Numerical

Let S be the mirror image of the point Q(1, 3, 4) with respect to the plane 2x − y + z + 3 = 0 and let R(3, 5, γ) be a point of this plane. Then the square of the length of the line segment SR is _____.

Open solution

Q402. JEE Main 2021 (Online) 26th August Evening Shift · Numerical

Let Q be the foot of the perpendicular from the point P(7, −2, 13) on the plane containing the lines $\frac{x+1}{6} = \frac{y-1}{7} = \frac{z-3}{8}$ and $\frac{x-1}{3} = \frac{y-2}{5} = \frac{z-3}{7}$. Then $(PQ)^2$, is equal to _____.

Open solution

Q403. JEE Main 2021 (Online) 26th August Morning Shift · Numerical

Let the line L be the projection of the line $\frac{x-1}{2} = \frac{y-3}{1} = \frac{z-4}{2}$ in the plane x − 2y − z = 3. If d is the distance of the point (0, 0, 6) from L, then d^2 is equal to _____.

Open solution

Q404. JEE Main 2021 (Online) 27th July Evening Shift · Numerical

The distance of the point P(3, 4, 4) from the point of intersection of the line joining the points Q(3, −4, −5) and R(2, −3, 1) and the plane 2x + y + z = 7, is equal to _____.

Open solution

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Q405. JEE Main 2021 (Online) 27th July Morning Shift · **Numerical**

Let a plane P pass through the point (3, 7, -7) and contain the line, $\frac{x-2}{-3} = \frac{y-3}{2} = \frac{z+2}{1}$. If distance of the plane P from the origin is d, then d^2 is equal to _____.

Open solution

Q406. JEE Main 2021 (Online) 25th July Evening Shift · **Numerical**

If the lines $\frac{x-k}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and $\frac{x+1}{3} = \frac{y+2}{2} = \frac{z+3}{1}$ are co-planar, then the value of k is _____.

Open solution

Q407. JEE Main 2021 (Online) 20th July Morning Shift · **Numerical**

Let P be a plane passing through the points (1, 0, 1), (1, -2, 1) and (0, 1, -2). Let a vector $\vec{a} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k}$ be such that $\vec{a}$ is parallel to the plane P, perpendicular to $(\hat{i} + 2\hat{j} + 3\hat{k})$ and $\vec{a}.\ (\hat{i} + \hat{j} + 2\hat{k}) = 2$, then $(\alpha - \beta + \gamma)^2$ equals _____.

Open solution

Q408. JEE Main 2021 (Online) 18th March Evening Shift · **Numerical**

Let the mirror image of the point (1, 3, a) with respect to the plane $\vec{r}.\left(2\hat{i} - \hat{j} + \hat{k}\right) - b = 0$ be (-3, 5, 2). Then, the value of | a + b | is equal to _____.

Open solution

Q409. JEE Main 2021 (Online) 18th March Evening Shift · **Numerical**

Let P be a plane containing the line $\frac{x-1}{3} = \frac{y+6}{4} = \frac{z+5}{2}$ and parallel to the line $\frac{x-1}{4} = \frac{y-2}{-3} = \frac{z+5}{7}$. If the point (1, -1, α) lies on the plane P, then the value of $|5\alpha|$ is equal to _____.

Open solution

Q410. JEE Main 2021 (Online) 18th March Morning Shift · **Numerical**

Let the plane ax + by + cz + d = 0 bisect the line joining the points (4, -3, 1) and (2, 3, -5) at the right angles. If a, b, c, d are integers, then the minimum value of ($a^2 + b^2 + c^2 + d^2$) is _____.

Open solution

Q411. JEE Main 2021 (Online) 18th March Morning Shift · **Numerical**

The equation of the planes parallel to the plane x $-$ 2y + 2z $-$ 3 = 0 which are at unit distance from the point (1, 2, 3) is ax + by + cz + d = 0. If (b $-$ d) = k(c $-$ a), then the positive value of k is :

Open solution

Q412. JEE Main 2021 (Online) 17th March Evening Shift · **Numerical**

Let P be an arbitrary point having sum of the squares of the distances from the planes x + y + z = 0, lx $-$ nz = 0 and x $-$ 2y + z = 0, equal to 9. If the locus of the point P is $x^2 + y^2 + z^2 = 9$, then the value of l $-$ n is equal to _____.

Open solution

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Q413. JEE Main 2021 (Online) 17th March Morning Shift · **Numerical**

If the equation of the plane passing through the line of intersection of the planes $2x - 7y + 4z - 3 = 0$, $3x - 5y + 4z + 11 = 0$ and the point $(-2, 1, 3)$ is $ax + by + cz - 7 = 0$, then the value of $2a + b + c - 7$ is _____.

Open solution

Q414. JEE Main 2021 (Online) 16th March Evening Shift · **Numerical**

If the distance of the point $(1, -2, 3)$ from the plane $x + 2y - 3z + 10 = 0$ measured parallel to the line, $\frac{x-1}{3} = \frac{2-y}{m} = \frac{z+3}{1}$ is $\sqrt{\frac{7}{2}}$, then the value of $|m|$ is equal to _____.

Open solution

Q415. JEE Main 2021 (Online) 26th February Morning Shift · **Numerical**

Let $(\lambda, 2, 1)$ be a point on the plane which passes through the point $(4, -2, 2)$. If the plane is perpendicular to the line joining the points $(-2, -21, 29)$ and $(-1, -16, 23)$, then $\left(\frac{\lambda}{11}\right)^2 - \frac{4\lambda}{11} - 4$ is equal to _____.

Open solution

Q416. JEE Main 2020 (Online) 4th September Morning Slot · **Numerical**

If the equation of a plane P, passing through the intersection of the planes,
$x + 4y - z + 7 = 0$ and $3x + y + 5z = 8$ is $ax + by + 6z = 15$ for some $a, b \in R$, then the distance of the point (3, 2, -1) from the plane P is.....

Open solution

Q417. JEE Main 2020 (Online) 3rd September Evening Slot · **Numerical**

Let a plane P contain two lines
$\vec{r} = \hat{i} + \lambda\left(\hat{i} + \hat{j}\right), \lambda \in R$ and
$\vec{r} = -\hat{j} + \mu\left(\hat{j} - \hat{k}\right), \mu \in R$
If $Q(\alpha, \beta, \gamma)$ is the foot of the perpendicular drawn from the point M(1, 0, 1) to P, then $3(\alpha + \beta + \gamma)$ equals _____.

Open solution

Q418. JEE Main 2020 (Online) 9th January Evening Slot · **Numerical**

If the distance between the plane, $23x - 10y - 2z + 48 = 0$ and the plane

containing the lines $\frac{x+1}{2} = \frac{y-3}{4} = \frac{z+1}{3}$

and $\frac{x+3}{2} = \frac{y+2}{6} = \frac{z-1}{\lambda}$ $(\lambda \in R)$

is equal to $\frac{k}{\sqrt{633}}$, then k is equal to _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **C**	2. **C**	3. **C**	4. **B**	5. **C**
6. **C**	7. **C**	8. **C**	9. **B**	10. **A**
11. **B**	12. **B**	13. **A**	14. **C**	15. **C**
16. **B**	17. **C**	18. **A**	19. **A**	20. **C**
21. **B**	22. **D**	23. **A**	24. **A**	25. **D**
26. **A**	27. **A**	28. **B**	29. **B**	30. **B**
31. **A**	32. **C**	33. **D**	34. **D**	35. **D**
36. **C**	37. **C**	38. **D**	39. **C**	40. **A**
41. **A**	42. **A**	43. **A**	44. **A**	45. **D**
46. **B**	47. **A**	48. **C**	49. **A**	50. **C**
51. **B**	52. **D**	53. **C**	54. **D**	55. **D**
56. **C**	57. **C**	58. **C**	59. **A**	60. **B**
61. **B**	62. **A**	63. **B**	64. **A**	65. **B**
66. **B**	67. **D**	68. **D**	69. **D**	70. **A**
71. **A**	72. **B**	73. **A**	74. **D**	75. **B**
76. **C**	77. **A**	78. **A**	79. **C**	80. **C**
81. **D**	82. **C**	83. **C**	84. **D**	85. **D**
86. **A**	87. **B**	88. **C**	89. **D**	90. **A**
91. **C**	92. **D**	93. **A**	94. **C**	95. **C**
96. **D**	97. **B**	98. **B**	99. **B**	100. **A**
101. **A**	102. **A**	103. **A**	104. **B**	105. **A**
106. **C**	107. **A**	108. **A**	109. **A**	110. **A**
111. **A**	112. **C**	113. **D**	114. **A**	115. **B**
116. **A**	117. **C**	118. **B**	119. **D**	120. **C**
121. **A**	122. **D**	123. **D**	124. **A**	125. **C**
126. **B**	127. **C**	128. **B**	129. **C**	130. **B**
131. **C**	132. **C**	133. **C**	134. **C**	135. **B**
136. **C**	137. **C**	138. **A**	139. **C**	140. **C**
141. **B**	142. **D**	143. **C**	144. **A**	145. **B**
146. **B**	147. **C**	148. **A**	149. **B**	150. **C**

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151. **A**	152. **B**	153. **C**	154. **B**	155. **D**
156. **A**	157. **D**			

Numerical

158. **4**	159. **162**	160. **170**	161. **6**	162. **3**
163. **56**	164. **957**	165. **216**	166. **62**	167. **11**
168. **43**	169. **13**	170. **25**	171. **6**	172. **21**
173. **22**	174. **12**	175. **196**	176. **16**	177. **9**
178. **65**	179. **108**	180. **5**	181. **18**	182. **158**
183. **9**	184. **18**	185. **384**	186. **14**	187. **75**
188. **450**	189. **153**	190. **6**	191. **125**	192. **51**
193. **84**	194. **2**	195. **6**	196. **44**	197. **1**
198. **8**	199. **4**			

Out of syllabus

200. **C**	201. **D**	202. **A**	203. **D**	204. **C**
205. **A**	206. **D**	207. **B**	208. **A**	209. **A**
210. **D**	211. **C**	212. **B**	213. **D**	214. **C**
215. **A**	216. **B**	217. **B**	218. **B**	219. **C**
220. **A**	221. **D**	222. **B**	223. **A**	224. **A**
225. **D**	226. **C**	227. **A**	228. **A**	229. **A**
230. **D**	231. **B**	232. **D**	233. **B**	234. **B**
235. **B**	236. **D**	237. **B**	238. **D**	239. **B**
240. **B**	241. **B**	242. **D**	243. **C**	244. **C**
245. **C**	246. **B**	247. **B**	248. **C**	249. **D**
250. **B**	251. **A**	252. **B**	253. **A**	254. **C**
255. **B**	256. **A**	257. **C**	258. **B**	259. **C**
260. **B**	261. **D**	262. **A**	263. **A**	264. **D**
265. **D**	266. **C**	267. **B**	268. **D**	269. **D**
270. **C**	271. **B**	272. **D**	273. **B**	274. **B**
275. **A**	276. **A**	277. **D**	278. **C**	279. **A**
280. **C**	281. **C**	282. **D**	283. **B**	284. **D**
285. **D**	286. **C**	287. **D**	288. **B**	289. **B**
290. **D**	291. **D**	292. **C**	293. **B**	294. **A**

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295. **B**	296. **D**	297. **D**	298. **C**	299. **A**
300. **B**	301. **C**	302. **C**	303. **A**	304. **C**
305. **C**	306. **C**	307. **D**	308. **D**	309. **B**
310. **C**	311. **B**	312. **B**	313. **C**	314. **C**
315. **C**	316. **C**	317. **B**	318. **C**	319. **A**
320. **A**	321. **B**	322. **B**	323. **C**	324. **A**
325. **A**	326. **D**	327. **A**	328. **A**	329. **C**
330. **C**	331. **C**	332. **A**	333. **A**	334. **D**
335. **A**	336. **B**	337. **D**	338. **B**	339. **A**
340. **A**	341. **B**	342. **C**	343. **B**	344. **C**
345. **B**	346. **C**	347. **C**	348. **C**	349. **C**
350. **D**	351. **C**	352. **D**	353. **A**	354. **B**
355. **C**	356. **C**	357. **A**	358. **D**	359. **A**
360. **A**	361. **C**	362. **D**	363. **A**	364. **C**
365. **B**	366. **B**	367. **C**	368. **A**	369. **D**
370. **D**	371. **A**	372. **A**	373. **B**	374. **26**
375. **15**	376. **288**	377. **11**	378. **7**	379. **11**
380. **9**	381. **594**	382. **10**	383. **6**	384. **180**
385. **9**	386. **15**	387. **315**	388. **355**	389. **9**
390. **10**	391. **12**	392. **2**	393. **125**	394. **3**
395. **26**	396. **28**	397. **137**	398. **5**	399. **7**
400. **61**	401. **72**	402. **96**	403. **26**	404. **7**
405. **3**	406. **1**	407. **81**	408. **1**	409. **38**
410. **28**	411. **4**	412. **0**	413. **4**	414. **2**
415. **8**	416. **3**	417. **5**	418. **3**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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